

# PROJECT **SILVA**: Spatial Intelligence for Living Vegetation & Agroforestry

An Integrated 1890 Land-Grant Platform for 3D Digital Twins, Remote Sensing, Forest Bioeconomy, and Robotic Agricultural Extension

| FUNDING AGENCY          | SOLICITATION / NOFO   | TOTAL REQUEST         | SUBMISSION ROUTE           |
|-------------------------|-----------------------|-----------------------|----------------------------|
| USDA NIFA (CFDA 10.216) | USDA-NIFA-CBGP-011698 | \$750,000 (36 Months) | NC A&T DOR Internal Portal |

**Strategic Project Framework (Consultation with Department Chair Dr. Niroj Aryal):** This proposal focuses on *technological capacity building, cross-college research infrastructure, enterprise spatial computing, and professional industry credentials*—establishing permanent institutional assets, accredited graduate degree pathways, and statewide extension capabilities across North Carolina A&T.

**Project Executive Summary & Capacity Mission:** This 36-month, \$750,000 Integrated Project establishes the *Project SILVA (Spatial Intelligence for Living Vegetation & Agroforestry)* across three Colleges at North Carolina A&T State University (CAES, CoST, COE) in direct partnership with Cooperative Extension at the Small Farm Research and Innovation Center (SFRIC). Rather than subsidizing routine departmental operations, this initiative creates **enduring institutional capacity** by: (1) establishing the permanent **Spatial Design Center & Computing Lab Space** equipped with high-performance GPU nodes and deploying the enterprise **SILVA Spatial Computing Ecosystem** (advancing the **AVA** AI design and SITES v2 rating engine integrated with **GeoScope GIS**); (2) building the technical and curricular scaffolding for **two new graduate degree programs**—the professional **Accelerated 4+1 Master of Landscape Architecture (BSLA + MLA)** and a multidisciplinary **M.S. in Applied Environmental Planning**; (3) establishing permanent biological field infrastructure through **The Living Fungal Forest Laboratory** (pecan truffle *Tuber lyonii* inoculations and fungal bioeconomy plots with 15–20 year research horizons); (4) credentialing **45+ historically underrepresented students** with verified national certifications (*NRCS Conservation Planning, SITES AP, ABCD Leadership*); and (5) converting SFRIC into an active **Living Extension Lab** with autonomous quadruped robotics and 3D digital farm twins for underserved landowners.

01. INTEGRATED LAND-GRANT FRAMEWORK

\$750,000 Total Allocation

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <div>TEACHING TRACK (PRIMARY)</div> <div>Spatial Studio &amp; Digital Twin Curriculum</div> <div>\$350,000 (46.7%)</div> <div>Builds the high-performance Digital Twin Studio (GeoScope + Cesium/LiDAR 3D spatial analysis) and embeds verifiable industry credentials (NRCS Conservation Planning, SITES AP, ABCD Community Development) into core environmental and design curricula.</div> <div>Leads: W. Chris Harrison (PI &amp; 3D Modeling Co-Lead), Dr. M. Chakraborty (Co-PI &amp; 3D Systems Co-Lead), Dr. J. Yang (Co-PI), J. Kuehner (Co-Inv), Dr. C. Warren English (Co-PI), Dr. P. Spence (Co-PI)</div> | <div>RESEARCH TRACK (SECONDARY)</div> <div>Canopy Sensing &amp; Forest Bioeconomy</div> <div>\$225,000 (30.0%)</div> <div>Deploys drone-based LiDAR, multispectral canopy imaging, 3D photogrammetric mesh modeling, and soil biome analytics to measure carbon sequestration, woodland polycultures, and high-value non-timber forest products (pecan truffle inoculation, medicinal fungi).</div> <div>Leads: W. Chris Harrison (PI &amp; 3D Modeling Co-Lead), Dr. M. Chakraborty (Co-PI &amp; 3D Systems Co-Lead), Dr. F. N. Anike (Co-PI), Dr. L. Hashemi Beni (Co-PI)</div> | <div>EXTENSION TRACK (TERTIARY)</div> <div>SFRIC Living Lab &amp; Multi-Tier Scouting</div> <div>\$175,000 (23.3%)</div> <div>Operationalizes the SFRIC Living Extension Lab utilizing multi-tier scouting —aerial drones for tree canopy health and ground robotics for understory/trunk diagnostics—paired with 3D digital farm twins and field workshops for small farmers.</div> <div>Leads: C. Hopper (SFRIC Director), Dr. H. Talton (Extension Specialist &amp; IPM Lead), W. Chris Harrison (Lead PI), Dr. M. Chakraborty (Robotics Co-Lead)</div> |
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**Strategic Project Identity Mandate (Senior Faculty Advisor Dr. Omoanghe Isikhuemhen):** During executive review sessions, Senior Faculty Advisor Dr. Omon Isikhuemhen advised against generic, bureaucratic institutional titles (such as "LL-DAL"). To command instant federal reviewer attention, inspire student cohorts, and create a lasting public brand, Dr. Omon mandated adopting a **human, nature-rooted, and highly memorable acronym**—matching the signature persona design established by Project Director W. Chris Harrison across platforms like **AVA**, **EMMA**, and **GeoScope**. In response, leadership has selected **Project SILVA** as the primary grant identity, evaluated alongside two other top-tier candidates.

PRIMARY ADOPTED IDENTITY

PROJECT SILVA

Spatial Intelligence for Living Vegetation & Agroforestry

**Biological Etymology:** Derived from the classical Latin *silva* ("forest / woodland"), directly honoring Dr. Omon and Dr. Anike's agroforestry and mycology research.

**Grant Subtitle:** *An Integrated 1890 Land-Grant Platform for 3D Digital Twins, Woodland Bioeconomy, and Precision Drone-Robotic Extension.*

**Reviewer Vibe:** Classical scientific authority. Rolls off the tongue like AVA while establishing an elite research brand.

STRATEGIC CANDIDATE 2

PROJECT CANOPY

Computational Agroforestry Network & Observation Platform for Yield

**Overstory & Aerial Sensing:** Emphasizes tree crown health, aerial drone LiDAR canopy penetration (Dr. Hashemi-Beni), and the microclimatic protective layer shielding understory polycultures.

**Grant Subtitle:** *Linking 3D Landscape Architecture Studios, Forest Bioeconomy, and Drone-Assisted Field Extension.*

**Reviewer Vibe:** Elevated, protective, and expansive. Natural metaphorical umbrella representing institutional Land-Grant capacity.

STRATEGIC CANDIDATE 3

PROJECT ROOTS

Resilient Observation & Technology for Soils

**Biological & HBCU Anchor:** Anchors the biological root-mycorrhizal fungal interface while celebrating 1890 HBCU land-grant legacy and community grounding.

**Grant Subtitle:** *An Integrated Land-Grant Initiative for Digital Landscape Studios, Mycorrhizal Bioeconomy, and Small Farm Extension.*

**Reviewer Vibe:** Emotionally resonant and purpose-driven. Direct alignment with USDA equity and soil health priorities.

COMPARATIVE IDENTITY MATRIX: STRATEGIC EVALUATION FOR FEDERAL REVIEW

| IDENTITY OPTION | FULL STRATEGIC EXPANSION                                            | ACADEMIC & BIOLOGICAL DOMAIN                                                                | REVIEWER PSYCHOLOGY & VIBE                                                 | ADOPTION STATUS    |
|-----------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|--------------------|
| PROJECT SILVA   | Spatial Intelligence for Living Vegetation & Agroforestry           | Forest canopy, silviculture, drone LiDAR, and non-timber forest mycology.                   | High academic dignity; commands national center prestige.                  | ★ ADOPTED          |
| PROJECT CANOPY  | Computational Agroforestry Network & Observation Platform for Yield | Tree crown health, drone LiDAR canopy penetration, microclimatic modeling, and shade crops. | Elevated, expansive; natural protective umbrella for agroforestry systems. | CANOPY ALTERNATE   |
| PROJECT ROOTS   | Resilient Observation & Technology for Soils                        | Soil health, truffle mycorrhizae, carbon sequestration, and farmer equity.                  | Deeply grounded in HBCU Land-Grant mission and soil biology.               | HERITAGE ALTERNATE |

**Formal Brand Adoption:** Per team consensus, **Project SILVA** is officially adopted as the primary grant identity, seamlessly retiring the working acronym "LL-DAL" across Cayuse institutional routing and the full narrative submission.

02. DEFINING INSTITUTIONAL CAPACITY BUILDING & THE GRADUATE DEGREE BRIDGE

Sustainable Institutional Transformation

**USDA NIFA Capacity Building Standard (7 U.S.C. § 3152 & NOFO USDA-NIFA-CBGP-011698):** As highlighted in consultation with Department Chair Dr. Niroj Aryal, federal review panels explicitly disqualify applications that appear to subsidize routine faculty effort, everyday classroom supplies, or incremental course revisions. Under federal guidelines, **Capacity Building** requires establishing *permanent institutional assets, accredited degree pathways, cross-college research pipelines, and statewide field infrastructure* that fundamentally transform an 1890 Land-Grant institution's competitive capabilities long after the 36-month grant period ends.

REVIEWER EVALUATION BENCHMARK: ROUTINE DEPARTMENTAL ACTIVITY VS. PERMANENT CAPACITY TRANSFORMATION

| DIMENSION                        | ROUTINE / NON-CAPACITY PROPOSAL (DISQUALIFIED)                                                                                       | PROJECT SILVA CAPACITY TRANSFORMATION (FUNDED & ENDURING)                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Academic Degree Infrastructure   | Teaching existing undergraduate lecture courses; incremental syllabus updates; no new degree or professional accreditation creation. | <b>Graduate Elevation Bridge:</b> Curricular, laboratory, and technological scaffolding laying the essential groundwork to establish the <b>Accelerated 4+1 Master of Landscape Architecture (BSLA + MLA)</b> (first HBCU LAAB-accredited MLA in modern decades) and a <b>Multidisciplinary M.S. in Applied Environmental Planning</b> .                                                                                                         |
| Computing & Spatial Hardware     | Purchasing off-the-shelf consumer laptops or temporary software licenses that expire at grant termination.                           | <b>Permanent Lab Asset &amp; Enterprise Toolset (AVA + GeoScope):</b> Establishes the permanent <b>Spatial Design Center &amp; Computing Lab Space</b> (6 GPU workstations, VR suites, drone docks) and deploys the dual-engine spatial computing architecture: the <b>AVA (Adaptive Visualization Assistant)</b> AI design and <b>SITES v2</b> ecological rating engine coupled with the <b>GeoScope</b> open GIS intelligence platform.        |
| Research & Biological Assets     | Single-investigator summer research projects conducted in departmental isolation with temporary findings.                            | <b>Living Fungal Forest Laboratory &amp; Tri-College Core:</b> Establishes permanent biological field infrastructure through the <b>Living Fungal Forest Laboratory</b> (pecan truffle <i>Tuber lyonii</i> inoculated rootzones and medicinal fungal polycultures at SFRIC and research woodlots with 15–20 year research lifespans), coupled with drone LiDAR point cloud archives and a multi-agency grant engine linking CAES, CoST, and COE. |
| Student Outcomes & Human Capital | Standard course grades; generic undergraduate diplomas without external industry validation.                                         | <b>Verifiable Industry Credentials:</b> <b>45+ historically underrepresented students</b> earn nationally recognized certifications ( <i>USDA NRCS Apprentice Conservation Planner (NRCS-CP-Apprentice)</i> , <i>SITES AP</i> , <i>ABCD Community Leadership</i> ), funneling diverse talent directly into federal agencies and premier firms.                                                                                                   |

| DIMENSION                | ROUTINE / NON-CAPACITY PROPOSAL (DISQUALIFIED)                                                                     | PROJECT SILVA CAPACITY TRANSFORMATION (FUNDED & ENDURING)                                                                                                                                                                              |
|--------------------------|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Extension & Field Assets | Passive PowerPoint seminars or webinars with no active field demonstration tech or landowner engagement.           | <b>Living Extension Lab:</b> Transforms the 100+ acre SFRIC facility into a permanent living lab featuring <b>autonomous quadruped ("robot dog") scouting</b> , digital farm twins, and recurring regional farmer training field days. |
| Post-Grant Longevity     | Project activity terminates when grant funds cease; equipment is abandoned or repurposed for routine office tasks. | <b>Institutionalized &amp; Sustainable:</b> Permanent physical lab space, institutionalized course syllabi, recurring graduate student tuition revenue, university-hosted web engines, and ongoing field research plots.               |

### 1. Digital & Spatial Computing Infrastructure (AVA + GeoScope Ecosystem)

Establishes the permanent **Living Landscapes & Spatial Digital Twin Design Center & Collaborative Lab Space** (within CAES / interdisciplinary academic facilities), equipped with 6 high-performance GPU rendering workstations, immersive VR simulation suites, and drone data docks. Deploys the enterprise dual-engine platform: advancing the Project Director's proven **AVA (Adaptive Visualization Assistant)** AI design and Sustainable SITES Initiative (SITES v2) 250-point ecological rating engine, seamlessly integrated with **GeoScope** for parcel-scale GIS, USGS 3DEP topography, SSURGO soils, and drone telemetry. *(Under USDA NIFA CBG Educational Technology provisions, funding explicitly supports customizing, expanding, and institutionalizing these advanced spatial tools for multi-user university access.)*

### 2. Permanent Biological Infrastructure: The Living Fungal Forest Laboratory

Creates permanent biological research infrastructure by establishing **The Living Fungal Forest Laboratory & Agroforestry Mycobiome Center** across NC A&T research woodlots and the 100+ acre SFRIC facility. Inoculating managed woodland and pecan orchard rootzones with the high-value native pecan truffle (*Tuber lyonii*) and cultivating medicinal fungal polycultures establishes an enduring field research asset with a 15–20+ year production and scientific horizon. Enables continuous student research in soil carbon flux, fungal biotechnology (Dr. Anike), 3D canopy microclimate modeling (Harrison & Dr. Chakraborty), and high-value non-timber crop viability for underserved landowners.

### 3. Human Capital & USDA Workforce Credentialing Capacity

Establishes an institutionalized credentialing pipeline where **45+ historically underrepresented students** graduate with nationally recognized credentials: *NRCS Conservation Planning Apprentice, SITES AP, and ABCD Community Development*. Faculty train-the-trainer workshops ensure internal certification delivery continues perpetually, directly supplying diverse talent to USDA agencies (NRCS, USFS, FSA) and premier planning firms.

### 4. Statewide Cooperative Extension & Living Lab Capacity

Permanently transforms the 100+ acre Small Farm Research and Innovation Center (SFRIC) into an active **Living Extension Lab** equipped with autonomous quadruped ("robot dog") scouting, field-scale parcel digital twins, and recurring farmer training infrastructure specifically serving small-scale, limited-resource, and minority North Carolina landowners.

## CURRICULAR ARCHITECTURE: HOW PROJECT SILVA DIRECTLY SUPPORTS & BUILDS UPON NC A&T'S NEWLY IMPLEMENTED BSLA CURRICULUM

A critical strength of this proposal is that it does *not* disrupt or replace NC A&T's newly implemented 4-year undergraduate curriculum. Instead, Project SILVA acts as an institutional accelerator—supplying cutting-edge computational hardware, open-source GIS tools, and industry credentials that directly enrich each academic cohort, laying the academic foundation for the planned **Accelerated 4+1 Master of Landscape Architecture (BSLA + MLA)**:

FRESHMAN YEAR • 34 TOTAL CREDITS (21 LA CREDITS • 13 GE CREDITS)

| First Semester (Fall) |                       | 16 Credits        | Second Semester (Spring) |                          | 18 Credits         |
|-----------------------|-----------------------|-------------------|--------------------------|--------------------------|--------------------|
| LDAR 103              | LA Discovery          | 3 cr Major Core   | LDAR 104                 | LA History               | 3 cr Major Core    |
| LDAR 147              | Design Foundations I  | 3 cr Studio 1     | LDAR 148                 | LA Fundamentals          | 3 cr Pre-req: 147  |
| LDAR 170              | LA Graphics I         | 3 cr BIM / CAD    | LDAR 171                 | LA Graphics II           | 3 cr Pre-req: 170  |
| ENG 100               | Ideas & Expressions I | 3 cr Gen Ed       | LDAR 181                 | Materials & Construction | 3 cr Pre-req: 170  |
| MATH 101              | Analytical Reasoning  | 3 cr Gen Ed       | ENG 101                  | Ideas & Expressions II   | 3 cr Gen Ed        |
| FRT 101               | University Survival   | 1 cr Orientation  | MATH 102                 | Analytical Reasoning II  | 3 cr Gen Ed        |
| Subtotal              |                       | 16 cr 9 LA / 7 GE | Subtotal                 |                          | 18 cr 12 LA / 6 GE |

SOPHOMORE YEAR • 31 TOTAL CREDITS (24 LA CREDITS • 7 GE CREDITS)

| First Semester (Fall) |                               | 16 Credits           | Second Semester (Spring) |                           | 15 Credits         |
|-----------------------|-------------------------------|----------------------|--------------------------|---------------------------|--------------------|
| LDAR 204              | Plant Materials               | 3 cr Pre-req: 148    | LDAR 244                 | Designing with Plants     | 3 cr Pre-req: 204  |
| LDAR 247              | Social Systems Studio         | 6 cr Pre-req: 148    | LDAR 248                 | Ecological Systems Studio | 6 cr Pre-req: 247  |
| SBS                   | Social/Behavioral Science     | 3 cr Gen Ed Elective | LDAR 249                 | Construction Documents    | 3 cr Pre-req: 181  |
| SR                    | Scientific Reasoning Elective | 4 cr Gen Ed (w/ Lab) | LDAR 255                 | Design Theory             | 3 cr Major Core    |
| Subtotal              |                               | 16 cr 9 LA / 7 GE    | Subtotal                 |                           | 15 cr 15 LA / 0 GE |

JUNIOR YEAR • 31 TOTAL CREDITS (19 LA CREDITS • 12 GE CREDITS)

| First Semester (Fall) |                           | 16 Credits                 | Second Semester (Spring) |                               | 15 Credits               |
|-----------------------|---------------------------|----------------------------|--------------------------|-------------------------------|--------------------------|
| LDAR 345              | Site Systems              | 3 cr Co-req: 347; Pre: 248 | LDAR 349                 | GIS in Environmental Design   | 3 cr Pre-req: 247        |
| LDAR 347              | Landform & Stormwater     | 6 cr Co-req: 345; Pre: 248 | LDAR 348                 | Master Planning/Field Study   | 6 cr Pre-req: 248        |
| LDAR 399              | Resume/Portfolio Seminar  | 1 cr Pre-req: 248          | SR                       | Scientific Reasoning Elective | 3 cr Gen Ed Elective     |
| AA                    | African American History  | 3 cr Gen Ed Elective       | Free Elective            |                               | 3 cr University Elective |
| GA                    | Global Awareness Elective | 3 cr Gen Ed Elective       | Subtotal                 |                               | 15 cr 9 LA / 6 GE        |
| Subtotal              |                           | 16 cr 10 LA / 6 GE         |                          |                               |                          |

SENIOR YEAR • 24 TOTAL CREDITS (21 LA CREDITS • 3 GE CREDITS)

| First Semester (Fall) |                               | 12 Credits         | Second Semester (Spring) |                         | 12 Credits           |
|-----------------------|-------------------------------|--------------------|--------------------------|-------------------------|----------------------|
| LDAR 440              | Design Proposal Writing       | 3 cr Pre-req: 348  | LDAR 442                 | Design in Practice      | 3 cr Pre-req: 447    |
| LDAR 447              | The Collaborative Studio      | 6 cr Pre-req: 348  | LDAR 448                 | Capstone Project Studio | 6 cr Pre-req: 447    |
| LDAR 455              | Current Issues & Topics in LA | 3 cr Pre-req: 348  | HFA                      | Humanities/Fine Arts    | 3 cr Gen Ed Elective |
| Subtotal              |                               | 12 cr 12 LA / 0 GE | Subtotal                 |                         | 12 cr 9 LA / 3 GE    |

| ACADEMIC LEVEL                                                  | NEWLY IMPLEMENTED CORE SYLLABI                                                                                                                                                                                                                                                                 | SILVA EMBEDDED TECHNOLOGY & CREDENTIALS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | FACULTY LEADERSHIP & STUDENT PROGRESSION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Year 1 (Freshman)</b><br>Foundations & Materials             | <b>LDAR 103</b> LA Discovery (3 cr)<br><b>LDAR 104</b> LA History (3 cr)<br><b>LDAR 147/148</b> Design Foundations I & II (6 cr)<br><b>LDAR 170/171</b> LA Graphics I & II / BIM (6 cr)<br><b>LDAR 181</b> Site Materials & Construction Methods (3 cr)                                        | <ul style="list-style-type: none"> <li>Spatial Design Center high-performance GPU nodes</li> <li>Generative AI concept synthesis &amp; visual computing</li> <li><b>LDAR 181 Technical Materials Mastery:</b> 15-module hardscape assemblies, permeable paving (PICP), concrete joints, subgrade drainage, and structural joinery catalog</li> <li>Vectorworks Landmark BIM &amp; CAD digital workflow integration</li> </ul>                                                                                                                 | <b>W. Chris Harrison (Freshman Cohort Lead):</b><br>Harrison leads foundational design studios (LDAR 147/148, 170/171) and curricular authoring for <b>LDAR 181 Site Materials</b> , establishing hand sketching discipline, material assembly rigor, and early transition into Spatial Design Center GPU workstations.                                                                                                                                                                                                                             |
| <b>Year 2 (Sophomore)</b><br>Ecological Systems & Detailing     | <b>LDAR 204</b> Landscape Plants (3 cr)<br><b>LDAR 244</b> Designing with Plants (3 cr)<br><b>LDAR 247</b> Social Systems Studio (6 cr)<br><b>LDAR 248</b> Ecological Systems Studio (6 cr)<br><b>LDAR 249</b> Construction Documents (3 cr)<br><b>LDAR 255</b> Landscape Design Theory (3 cr) | <ul style="list-style-type: none"> <li><b>Environmental Evaluation &amp; NRCS CPA-52:</b> Rigorous baseline ecological assessment, soil infiltration testing, watershed catchment runoff analysis, and resource inventory embedded in LDAR 248</li> <li><b>SITES AP Pathway:</b> 250-point sustainable site performance auditing via the AVA engine</li> <li><b>ABCD Certification:</b> Asset-Based Community Development field methods with local communities</li> <li>Infiltration, soil moisture &amp; FLIR thermal sensor kits</li> </ul> | <b>Dallas Bretzman (Sophomore Cohort Lead) &amp; Dr. Porché Spence:</b><br>Bretzman directs the sophomore studio sequence (LDAR 247/248), embedding empirical <b>environmental evaluation</b> into <b>LDAR 248 Ecological Systems Studio</b> in direct collaboration with Dr. Spence (Soil Biogeochemistry). Students conduct field evaluations at the SFRIC living lab (Charlie Hopper) and The Living Fungal Forest Laboratory (Dr. Anike & Dr. Isikhuemhen), grounding ecological theory in physical Piedmont agricultural and woodland systems. |
| <b>Year 3 (Junior)</b><br>Site Engineering & Master Planning    | <b>LDAR 345</b> Planting Design & Site Systems (3 cr)<br><b>LDAR 347</b> Landform & Stormwater Studio (6 cr)<br><b>LDAR 348</b> Master Planning Studio (6 cr)<br><b>LDAR 349</b> GIS Applications in Planning (3 cr)<br><b>LDAR 399</b> Portfolio Seminar (1 cr)                               | <ul style="list-style-type: none"> <li><b>GeoScope 3DEP Integration:</b> Real-time USGS 2-ft contours for digital grading &amp; cut/fill calculations</li> <li>SSURGO soil hydrology &amp; FEMA flood hazard overlays</li> <li><b>Master Construction Detailing Exemplar Library:</b> Structural bioretention and permeable paving details</li> </ul>                                                                                                                                                                                         | <b>Dr. Jun Yang (Junior Cohort Lead):</b><br>Dr. Yang leads intermediate master planning and site systems studios (LDAR 345/347/348), integrating advanced generative AI concept synthesis, spatial cognition metrics, and terrain modeling, supported by her longitudinal IRB-approved research study.                                                                                                                                                                                                                                             |
| <b>Year 4 (Senior BSLA)</b><br>Professional Practice & Capstone | <b>LDAR 440</b> Design Proposal Writing (3 cr)<br><b>LDAR 442</b> Design in Practice (3 cr)<br><b>LDAR 447</b> Collaborative Studio (6 cr)<br><b>LDAR 448</b> Senior Capstone Studio (6 cr)<br><b>LDAR 455</b> Current Issues in LA (3 cr)                                                     | <ul style="list-style-type: none"> <li><b>Interactive 3D Digital Twins:</b> High-fidelity drone photogrammetry meshes of farm &amp; urban sites</li> <li><b>USDA NRCS Apprentice Conservation Planner (NRCS-CP-Apprentice):</b> 9-step conservation plan certification (Dr. Spence)</li> <li><b>LDAR 442 Mastery:</b> Professional practice, contracts, office operations, and licensure preparation</li> </ul>                                                                                                                               | <b>Jeffrey A. Kuehner (Senior Cohort Lead) &amp; W. Chris Harrison:</b><br>Kuehner directs senior capstones (LDAR 447/448), <b>LDAR 442 Design in Practice</b> , LDAR 440 Design Proposal Writing, and LDAR 455 Current Issues in LA, bringing 15+ years of professional environmental design and studio practice to portfolio reviews. Harrison & Dr. Chakraborty pair to guide parcel-scale 3D digital twins.                                                                                                                                     |

| ACADEMIC LEVEL                                                            | NEWLY IMPLEMENTED CORE SYLLABI                                                                                                                                                                                                          | SILVA EMBEDDED TECHNOLOGY & CREDENTIALS                                                                                                                                                                                                                                                                                                                                                     | FACULTY LEADERSHIP & STUDENT PROGRESSION                                                                                                                                                                                                                             |
|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Year 5 (+1 Graduate Year)</b><br><b>Master of Landscape Arch (MLA)</b> | <b>LDAR 601</b> Advanced Spatial AI Studio<br><b>LDAR 603</b> Ecological Engineering Lab<br><b>LDAR 605</b> Geospatial Research Methods<br><b>LDAR 611</b> Master's Terminal Thesis<br><b>LDAR 615</b> Executive Licensure & Leadership | <ul style="list-style-type: none"> <li>• 4D environmental simulation (drone LiDAR + robotic telemetry)</li> <li>• Advanced structural civil details &amp; carbon flux modeling</li> <li>• L.A.R.E. Licensure Masterclass &amp; multi-agency federal contracting (advancing beyond undergraduate LDAR 442 Design in Practice)</li> <li>• Peer-reviewed journal manuscript defense</li> </ul> | <b>Graduate Faculty Leadership Core:</b><br>Harrison, Dr. Yang, Kuehner, Dr. Chakraborty, and Dr. Beni deliver advanced graduate studios meeting full <b>Landscape Architectural Accreditation Board (LAAB)</b> standards—establishing the first 4+1 MLA at an HBCU. |


**The Graduate Elevation Bridge: Scaffolding the MLA & Multidisciplinary M.S. Degree**

**LONG-TERM INSTITUTIONAL IMPACT**

North Carolina A&T currently offers the Bachelor of Science in Landscape Architecture (BSLA). To elevate the university's academic profile and meet soaring national demand for minority environmental leaders, this Capacity Building Grant serves as the direct technological and curricular incubator for **two planned graduate degree programs**:

**1. Accelerated 4+1 Master of Landscape Architecture (BSLA + MLA)**

Lays the essential curricular, computational, and accreditation groundwork to establish North Carolina A&T's first-professional graduate degree required for professional licensure. The digital twin design center, high-performance spatial computing hardware, and SITES v2 credentialing modules developed under this grant constitute the exact curricular and technical infrastructure required for **Landscape Architectural Accreditation Board (LAAB)** graduate accreditation feasibility—positioning NC A&T to become the national flagship for minority graduate landscape architects.

**2. Multidisciplinary M.S. in Applied Environmental Planning & Geospatial Science**

Constructs the interdisciplinary curriculum and research laboratory scaffolding for a future Master of Science jointly administered across **CAES, CoST, and COE**. Graduate students will specialize in precision agroforestry, remote sensing, and resilient watershed planning. The research plots (Dr. Anike), drone LiDAR pipelines (Dr. Beni & Dr. Chakraborty), and NRCS conservation frameworks (Dr. Spence) form the immediate thesis laboratories and graduate research positions.

*\*USDA NIFA CBG guidelines explicitly award high priority to capacity building proposals that construct the institutional scaffolding for new accredited graduate degrees at 1890 Land-Grant institutions.*



## LAAB Standard 5 Accreditation Audit: Instructional FTE Requirements for 4+1 Master's Expansion

LAAB STANDARD 5  
BENCHMARK

Under **Landscape Architectural Accreditation Board (LAAB) Standard 5 (Faculty)**, strict instructional Full-Time Equivalent (FTE) thresholds govern accredited landscape architecture programs:

### 1. Current Single Degree (BSLA Only): 5.0 FTE Minimum

- **4 Full-Time Faculty (4.0 FTE):** Harrison (Freshman Lead), Dallas Bretzman (Sophomore Lead), Dr. Jun Yang (Junior Lead), Jeffrey A. Kuehner (Senior Lead).
- **2 Adjunct Faculty (1.0 FTE):** Required to cover studio overflow sections and plant electives.
- **Total Current Capacity: 5.0 FTE** ✓ VERIFIED (Exactly meets LAAB minimum of 5.0 for an undergraduate program).

### 2. Dual Degree Expansion (BSLA + 4+1 MLA): 7.0 FTE Minimum

- **LAAB Dual-Program Mandate:** Academic units offering both bachelor's and master's degrees must maintain at least **7.0 instructional FTE**, with at least **5.0 full-time faculty** holding professional LA degrees.
- **Grant Institutional Leverage:** This \$750K grant provides the external funding, faculty release, and graduate tuition model enabling CAES leadership to secure the **5th full-time tenure-track faculty line** from the university.

## INSTITUTIONAL LONGEVITY & POST-GRANT SUSTAINMENT (YEARS 4 AND BEYOND)

**Facilities & Computing:** Spatial Design Center hardware and GeoScope server instances are permanently incorporated into CAES IT maintenance schedules and supported through academic lab tech allocations.

**Curricular Integration:** NRCS, SITES AP, and GeoScope modules are formally approved into required studio course syllabi across NC A&T's new curriculum (LDAR 103, 104, 147/148, 170/171, 181, 204, 244, 247/248, 249, 255, 345, 347/348, 349, 399, 440, 442, 447/448, 455), ensuring zero dependency on continued grant funds.

**Tuition & Grant Leverage:** The new MLA and M.S. degree programs generate recurring graduate tuition and state formula funding, while the cross-college research core actively competes for multi-million-dollar NSF, USDA AFRI, and federal awards.

**Institutional Governance & Personnel Classification:** To maintain strict compliance with NC A&T Division of Research (DOR) policy and federal grant mechanics, this proposal clearly distinguishes between **Co-Principal Investigators (Co-PIs)** and **Project Collaborators / Institutional Partners**:

- **Co-PIs (Faculty Academic Lines):** Hold tenure-track or tenured faculty appointments (CAES, CoST, COE), bear primary scientific/fiscal accountability, and are eligible for allowable summer research salary or academic effort release under USDA NIFA guidelines.
- **Collaborators & Facility Partners:** Charles "Charlie" Hopper serves as *Senior Institutional Collaborator & Living Lab Director*. As full-time 12-month administrative staff, Charlie provides vital SFRIC field acreage, facility access, and farmer networks as an in-kind institutional collaboration without complex staff salary encumbrances. Dr. Hannah Talton serves as designated Extension Co-Investigator leading field IPM diagnostics and grower demonstration workshops. Dr. Momtanu Chakraborty (COE) and a funded Computer Science & Engineering Student Assistant lead all robotics programming and ROS navigation, protecting Extension specialist capacity.

03. PROJECT GOVERNANCE: CO-PIs VS. PROJECT COLLABORATORS

Faculty Co-PIs & Institutional Collaborators

| Investigator & Title                                                                                                                                                                      | Discipline & College                                                                                         | Core Expertise & Grant Value                                                                                                                                                                                                                                                       | Role, Placement & Funding Status                                                                                                                                                                                                                                                                                                                                                                                                         |
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| <div>W. Chris Harrison</div> <div>Assistant Professor of Landscape Architecture</div> <div>PROJECT DIRECTOR • LEAD PI • 3D MODELING CO-LEAD</div>                                         | <div>Dept. of Natural Resources &amp; Environmental Design (NRED)</div> <div>CAES (Faculty)</div>            | <div>Spatial computing, interactive GIS interfaces (GeoScope / AVA), 3D digital twin design, photogrammetric reconstruction, resilient master planning, SITES v2 rating system.</div>                                                                                              | <div>Overall Project Director (Lead PI) &amp; 3D Modeling Co-Lead</div> <div>Directs Digital Twin Studio, platform architecture, DOR/NIFA compliance, and student studio synthesis.</div> <div>Directly paired with Dr. Momtanu Chakraborty to co-lead all 3D model creation, UAV photogrammetric mesh reconstruction, and interactive digital twin spatial simulations across AVA and GeoScope. (Faculty Effort Allocation)</div>       |
| <div>Dr. Jun Yang</div> <div>Assistant Professor of Landscape Architecture</div> <div>CO-PI • AI Pedagogy &amp; Design Cognition Lead</div>                                               | <div>Dept. of Natural Resources &amp; Environmental Design (NRED)</div> <div>CAES (Faculty)</div>            | <div>Junior studio leadership (LDAR 345/347/348/349), AI-assisted design pedagogy, spatial synthesis, design cognition &amp; behavioral analytics, land planning and environmental studio synthesis, CELA/EDRA research.</div>                                                     | <div>Co-PI (Junior Cohort Lead • AI Pedagogy &amp; Design Cognition)</div> <div>Directs junior design studios (LDAR 345 Site Systems, LDAR 347 Landform &amp; Stormwater, LDAR 348 Master Planning, and LDAR 349 GIS Applications); leads longitudinal IRB-approved AI spatial cognition assessment and co-authors pedagogy research papers. (Faculty Effort Allocation)</div>                                                           |
| <div>Jeffrey A. Kuehner, MLA, ASLA</div> <div>Full-Time Lecturer of Landscape Architecture</div> <div>CO-INVESTIGATOR • SENIOR CAPSTONE, PRACTICE &amp; CONSTRUCTION DETAILING LEAD</div> | <div>Dept. of Natural Resources &amp; Environmental Design (NRED)</div> <div>CAES (Faculty / Lecturer)</div> | <div>Senior capstone design studio leadership (LDAR 447/448), professional practice &amp; design proposal writing (LDAR 440/442), professional construction documentation (LDAR 249 building upon LDAR 181), constructible detailing, site engineering &amp; ASLA standards.</div> | <div>Co-Investigator (Senior Capstone Studio, Professional Practice &amp; Construction Detailing)</div> <div>Directs senior capstone design execution (LDAR 447/448), proposal writing (LDAR 440), design in practice (LDAR 442), and professional construction documents (LDAR 249 building upon LDAR 181); embeds the Master Construction Detailing Exemplar Library into senior capstone projects. (Academic Effort Allocation)</div> |

| INVESTIGATOR & TITLE                                                                                                                                          | DISCIPLINE & COLLEGE                                                                                                    | CORE EXPERTISE & GRANT VALUE                                                                                                                                                            | ROLE, PLACEMENT & FUNDING STATUS                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <b>Dr. Leila Hashemi Beni</b><br>Associate Professor & GEMS Director<br><b>CO-PI • GEOSPATIAL LEAD</b>                                                        | Dept. of Built Environment / GEMS Institute<br><b>College of Science &amp; Technology (Faculty)</b>                     | Aerial/terrestrial LiDAR, UAV remote sensing, photogrammetry, point cloud classification, deep learning for environmental feature extraction.                                           | <b>Co-PI (Geospatial &amp; Aerial Drone Lead)</b><br>Directs aerial drone multispectral/LiDAR overflights for tree canopy and forest crown diagnostics (identifying foliar stress from above), point cloud classification, and student training. <i>(Faculty Effort Allocation)</i>                                                                                                                                                               |
| <b>Dr. Felicia N. Anike</b><br>Research Scientist / Faculty • Fungal Biotechnology<br><b>CO-PI • BIOECONOMY LEAD</b>                                          | Dept. of Natural Resources & Environmental Design (Carver Hall)<br><b>CAES (Faculty / Research Scientist)</b>           | Mycology, non-timber forest products (NTFPs), truffle biotechnology ( <i>Tuber lyonii</i> ), fungal fermentation, crop residue valorization, agroforestry ecology.                      | <b>Co-PI (Forest Bioeconomy Research Lead)</b><br>Directs woodland field plots, fungal inoculation trials, canopy soil dynamics, and agroforestry carbon metrics. <i>(Succeeding Dr. Omon Isikhuemhen, who serves as Senior Project Advisor)</i>                                                                                                                                                                                                  |
| <b>Dr. Momtanu Chakraborty</b><br>Assistant Professor of Computational Data Science and Engineering<br><b>CO-PI • ROBOTICS, UAV &amp; 3D MODELING CO-LEAD</b> | Department of Computational Data Science and Engineering (McNair Hall)<br><b>College of Engineering (COE) (Faculty)</b> | UAV mission planning, multispectral sensor data pipelining, agricultural robotics hardware integration, AVA platform integration, 3D mesh reconstruction, point cloud-to-GIS data sync. | <b>Co-PI (Robotics, UAV Systems &amp; 3D Modeling Co-Lead)</b><br><b>Directly paired with Project Director W. Chris Harrison to co-lead all 3D modeling, UAV photogrammetry mesh pipelines, and interactive digital twin spatial engines.</b><br>Supervises the funded Computer Science / Engineering Student Assistant responsible for autonomous robot dog programming, ROS navigation, and field telemetry. <i>(Faculty Effort Allocation)</i> |
| <b>Dr. Porché L. Spence</b><br>Associate Professor of Environmental Studies<br><b>CO-PI • CONSERVATION LEAD</b>                                               | Dept. of Natural Resources & Environmental Design (Carver Hall)<br><b>CAES (Faculty)</b>                                | Soil health, urban watershed biogeochemistry, water quality diagnostics, ESA Diversity in Ecology Awardee, experiential student mentoring.                                              | <b>Co-PI (Conservation &amp; Credentials Lead)</b><br>Directs NRCS Conservation Planning alignment, soil/water lab analysis, and student research cohort mentoring. <i>(Faculty Effort Allocation)</i>                                                                                                                                                                                                                                            |

| INVESTIGATOR & TITLE                                                                                                                                                            | DISCIPLINE & COLLEGE                                                                                                    | CORE EXPERTISE & GRANT VALUE                                                                                                                                         | ROLE, PLACEMENT & FUNDING STATUS                                                                                                                                                                                                                                                                                                                                                                                             |
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| <b>Dr. Chastity Warren English</b><br>Associate Professor & Program Coordinator<br><div>CO-PI • EVALUATION LEAD</div>                                                           | Dept. of Agribusiness, Applied Economics and Agriscience Education (Carver Hall 251-A)<br><b>CAES (Faculty)</b>         | Agricultural pedagogy, learning outcome assessment, ACUE certified instruction, USDA evaluation frameworks, curriculum efficacy research.                            | <b>Co-PI (Project Evaluation Director)</b><br>Directs independent educational assessment, pre/post competency testing, logic model metrics, and pedagogy publications.<br><i>(Faculty Effort Allocation)</i>                                                                                                                                                                                                                 |
| <b>Charles "Charlie" Hopper</b><br>Director, Small Farm Research & Innovation Center<br><div>SENIOR COLLABORATOR • LIVING LAB DIRECTOR</div>                                    | SFRIC / Cooperative Extension<br><b>CAES Staff / Center Administration</b>                                              | Small farm administration, landowner engagement, ABCD (Asset-Based Community Development) methodologies, rural agricultural networks.                                | <b>Senior Institutional Collaborator</b><br>Provides SFRIC 100+ acre living lab demonstration site, coordinates farmer field day logistics, and leads ABCD community engagement. <i>(No Grant Salary Required — Administrative Facility Host)</i>                                                                                                                                                                            |
| <b>Dr. Hannah Talton, Ph.D., D.P.M.</b><br>Assistant Professor & Extension Specialist (IPM / Plant Pathology)<br><div>CO-INVESTIGATOR • EXTENSION &amp; FIELD DIAGNOSTICS</div> | Cooperative Extension / Agricultural Research<br><b>CAES Cooperative Extension (Faculty &amp; Extension Specialist)</b> | Plant pathology diagnostics, integrated pest management (IPM), insect vector control, understory/specialty crop scouting protocols, farmer demonstration field days. | <b>Extension Co-Investigator (Field Diagnostics &amp; IPM Outreach Lead)</b><br>Directs field diagnostic trials at SFRIC, disease/pest ground-truthing, grower demonstration field days, and Extension publications. <i>(Note: All robotics programming and autonomous navigation software are engineered by Dr. Chakraborty and a budgeted CS/Engineering Student Assistant, protecting Extension specialist capacity).</i> |
| <b>Dr. Omoanghe S. Isikhuemhen</b><br>Research Professor of Fungal Biotechnology & Agroforestry<br><div>SENIOR FACULTY PROJECT ADVISOR</div>                                    | Dept. of Natural Resources & Environmental Design (Carver Hall)<br><b>CAES (Faculty)</b>                                | Forest ecology, non-timber forest products, truffle biotechnology, international mycology networks, strategic project advisory.                                      | <b>Senior Faculty Project Advisor</b><br>Provides high-level scientific guidance and research mentorship to Dr. Anike and the project team without Co-PI administrative or fiscal encumbrances.                                                                                                                                                                                                                              |

04. INDIVIDUAL SCOPE OF WORK, WORKLOAD ALLOCATION & ACCOUNTABILITY MATRIX

Zero-Ambiguity Role Alignment

**Clear Workload & Responsibility Division:** To eliminate ambiguity, prevent faculty burnout, and establish transparent operational boundaries, this section defines the exact effort commitment, individual scope of work, tangible deliverables, and non-overlapping domain boundaries for every member of the 11-person project team across the three Land-Grant core areas.

| Investigator & Role                                                                                                                    | Grant Effort                                | Primary Operational Domain & Assigned Scope                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Individual Deliverables & Boundary Accountability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| <div>W. Chris Harrison, MLA</div> <div>PROJECT DIRECTOR • LEAD PI • 3D MODELING CO-LEAD</div> <div>Landscape Architecture • CAES</div> | 2.0 Person-Mos/yr Summer + 15% AY Direction | <b>Overall Project Leadership &amp; 3D Spatial Computing:</b> Directs \$750k fiscal/administrative compliance, annual NIFA REEport filings, and DOR liaison; oversees enterprise <b>GeoScope</b> digital twin deployment; supervises commissioning of the Spatial Design Center Lab; embeds SITES v2 credentialing into senior capstone studios; <b>co-leads the 3D Modeling &amp; Digital Twin Core paired directly with Dr. Momtanu Chakraborty</b> , generating interactive 3D site meshes, photogrammetric reconstructions, and web-based spatial twins across AVA and GeoScope. | <ul style="list-style-type: none"><li>GeoScope production platform live (Mo 9)</li><li>Spatial Design Center 6-GPU setup (Mo 6)</li><li>Interactive 3D Digital Twin Models of SFRIC &amp; Woodlots (Mo 12, paired with Chakraborty)</li><li>Annual NIFA Progress Reports (Mo 12, 24, 36)</li><li>Co-Lead Author on Paper 2 (Digital Twins) &amp; Co-Author on Papers 1, 3, 4, 5 in 5-Paper Suite</li></ul> <i>Boundary: Overall fiscal governance and co-leads all 3D modeling/digital twins paired with Chakraborty; does not run AI cognition testing (Yang) or site grading details (Kuehner).</i> |
| <div>Dr. Jun Yang, Ph.D.</div> <div>CO-PI • AI PEDAGOGY LEAD</div> <div>Landscape Architecture • CAES</div>                            | 1.0 Person-Mo/yr Summer + 10% AY Effort     | <b>Junior Studio Leadership, AI Pedagogy &amp; Design Cognition:</b> Directs junior design studios (LDAR 345 Site Systems, LDAR 347 Landform & Stormwater, LDAR 348 Master Planning Studio, and LDAR 349 GIS Applications); leads longitudinal IRB-approved cognition study tracking spatial synthesis, design velocity, and mental workload with generative AI; mentors intermediate design cohorts in computational master planning.                                                                                                                                               | <ul style="list-style-type: none"><li>AI studio modules &amp; syllabi updates (Mo 10)</li><li>IRB approval &amp; cognition dataset (Mo 14)</li><li>Lead Author on Paper 1 (AI Studio Pedagogy) in the 5-Paper Suite (CELA LRR / EDRA)</li><li>Co-Lead on SEC-02 &amp; SEC-03 proposal writing</li></ul> <i>Boundary: Sole faculty lead on design cognition research; does not manage hardware procurement (Harrison).</i>                                                                                                                                                                             |

| INVESTIGATOR & ROLE                                                                                                                              | GRANT EFFORT                                                      | PRIMARY OPERATIONAL DOMAIN & ASSIGNED SCOPE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | INDIVIDUAL DELIVERABLES & BOUNDARY ACCOUNTABILITY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <p><b>Jeffrey A. Kuehner, MLA, ASLA</b></p> <p><b>CO-INV • SENIOR PRACTICE &amp; DETAILING LEAD</b></p> <p>Lecturer of Landscape Arch • CAES</p> | <p><b>1.0 Person-Mo/yr Summer</b></p> <p>+ Studio Instruction</p> | <p><b>Senior Capstone Studio, Professional Practice &amp; Construction Detailing:</b></p> <p>Directs senior capstones (LDAR 447 The Collaborative Studio &amp; LDAR 448 Capstone Project Studio), design proposal writing (LDAR 440), design in practice (LDAR 442), and professional construction documentation (LDAR 249); develops the Master Construction Detailing Exemplar Library (linking LDAR 181 materials and GeoScope 3DEP topo data to constructible, buildable site details for senior capstone defense).</p> | <ul style="list-style-type: none"> <li>• Master Construction Detailing Exemplar Library (15-module hardscape, bioretention, and permeable paving catalog) (Mo 20)</li> <li>• Senior capstone technical construction documentation and buildability reviews (LDAR 447/448)</li> <li>• Integration of professional specifications and contract drafting into LDAR 249 &amp; LDAR 440/442</li> <li>• Co-Author on SEC-03 (Teaching Infrastructure) and SEC-07 (Facilities &amp; Detailing Labs)</li> </ul> <p><i>Boundary: Directs senior capstone design, practice, and construction documentation; does not direct junior AI cognitive testing (Yang) or server/drone systems (Harrison &amp; Chakraborty).</i></p> |
| <p><b>Dr. Chastity Warren English</b></p> <p><b>CO-PI • EVALUATION DIRECTOR</b></p> <p>Ag &amp; Extension Education • CAES</p>                   | <p><b>1.0 Person-Mo/yr Summer</b></p> <p>(Y1-Y3)</p>              | <p><b>Independent Project Evaluation:</b></p> <p>Directs formal, ACUE-grounded assessment plan across all teaching and extension modules; develops project Logic Model, pre/post competency rubrics, and student retention/graduation tracking; measures credential attainment across NRCS, SITES AP, and ABCD cohorts.</p>                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>• Project Logic Model &amp; Evaluation Plan (Mo 4)</li> <li>• Validated Pre/Post Assessment Instruments (Mo 6)</li> <li>• Annual Independent Evaluation Reports (Mo 12, 24, 36)</li> <li>• Co-Author on Paper 1 (Pedagogy) &amp; Paper 5 (Extension) in the 5-Paper Suite (NACTA/JOE)</li> </ul> <p><i>Boundary: Completely independent evaluation oversight; does not teach studios or manage lab hardware.</i></p>                                                                                                                                                                                                                                                        |
| <p><b>Dr. Porché L. Spence, Ph.D.</b></p> <p><b>CO-PI • CONSERVATION LEAD</b></p> <p>Environmental Studies • CAES</p>                            | <p><b>1.0 Person-Mo/yr Summer</b></p> <p>+ 10% AY Effort</p>      | <p><b>Conservation Credentials &amp; Soil Biogeochemistry:</b> Directs NRCS Conservation Planning Apprentice credential curriculum (USDA 9-step planning); leads lab analysis of soil health, soil organic carbon (SOC), and water quality from research woodlots; correlates ground-truth carbon with canopy remote sensing.</p>                                                                                                                                                                                           | <ul style="list-style-type: none"> <li>• NRCS Conservation Planning Course Modules (Mo 9)</li> <li>• Soil Biogeochemistry &amp; Carbon Stock Database (Mo 18, 30)</li> <li>• Mentoring 10 undergraduate student researchers</li> <li>• Lead Author on SEC-10 (Mentoring &amp; Diversity Plan)</li> </ul> <p><i>Boundary: Sole lead for NRCS credentialing &amp; soil carbon lab assays; does not run fungal inoculations (Anike) or drones (Beni).</i></p>                                                                                                                                                                                                                                                         |

| INVESTIGATOR & ROLE                                                                                                                                | GRANT EFFORT                                     | PRIMARY OPERATIONAL DOMAIN & ASSIGNED SCOPE                                                                                                                                                                                                                                                                                                                                                                                    | INDIVIDUAL DELIVERABLES & BOUNDARY ACCOUNTABILITY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| <b>Dr. Leila Hashemi Beni, Ph.D.</b><br><div>CO-PI • GEOSPATIAL LEAD</div> Built Environment / GEMS<br>• CoST                                      | <b>1.0 Person-Mo/yr Summer (Y1-Y3)</b>           | <b>LiDAR Remote Sensing &amp; Spatial Data Pipeline:</b> Directs UAV and terrestrial LiDAR data collection missions across research woodlots and SFRIC; manages point cloud classification (canopy height models, ground elevation DEMs); provides calibrated spatial data layers to the GeoScope engine.                                                                                                                      | <ul style="list-style-type: none"> <li>• Calibrated LiDAR 3D Point Cloud Archive (Mo 10)</li> <li>• Classified Digital Surface Models &amp; CHM (Mo 18)</li> <li>• Data Management Plan (DMP) Open Repository (Mo 6)</li> <li>• Lead Author on Paper 4 (LiDAR &amp; Soil Carbon) &amp; Co-Author on Paper 2 in 5-Paper Suite; Lead on SEC-07 &amp; SEC-09 <i>Boundary: Aerial/terrestrial LiDAR hardware &amp; point cloud processing; does not manage robot dog scouting (Talton/Hopper).</i></li> </ul>                                                                                                                                                                                                                             |
| <b>Dr. Felicia N. Anike, Ph.D.</b><br><div>CO-PI • BIOECONOMY LEAD</div> Fungal Biotechnology • CAES                                               | <b>1.5 Person-Mos/yr Summer (Y1-Y3 Research)</b> | <b>Forest Bioeconomy &amp; Fungal Biotechnology:</b> Directs agroforestry field trial establishment, fungal inoculation protocols (pecan truffle <i>Tuber lyonii</i> and medicinal fungi), mycorrhizal colonization tracking, soil enzyme assays, and non-timber forest crop viability in Carver Hall labs and field plots.                                                                                                    | <ul style="list-style-type: none"> <li>• Agroforestry Inoculation Protocols &amp; Field Setup (Mo 8)</li> <li>• Truffle Colonization &amp; Soil Microbiome Dataset (Mo 20, 32)</li> <li>• Co-Lead Author on SEC-04 (Research Plan)</li> <li>• Lead Author on Paper 3 (Fungal Forest &amp; Truffle Bioeconomy) in 5-Paper Suite (<i>Agroforestry Systems / Mycorrhiza</i>) <i>Boundary: Full scientific ownership of fungal/truffle research; succeeds Dr. Omon, who serves as Senior Project Advisor.</i></li> </ul>                                                                                                                                                                                                                  |
| <b>Dr. Momtanu Chakraborty, Ph.D.</b><br><div>CO-PI • ROBOTICS, UAV &amp; 3D MODELING CO-LEAD</div> Computational Data Science & Engineering • COE | <b>1.0 Person-Mo/yr Summer (Y1-Y3)</b>           | <b>3D Modeling, UAV Sensor Telemetry &amp; Simulation Engineering: Co-leads the 3D Modeling &amp; Digital Twin Core paired directly with Lead PI W. Chris Harrison;</b> engineers the UAV photogrammetry-to-3D-mesh processing pipeline, point-cloud conversion, and automated sensor telemetry into GeoScope and AVA; co-directs drone flight missions with CoST; provides engineering support for robotic quadruped sensing. | <ul style="list-style-type: none"> <li>• Multispectral UAV Flight Protocols &amp; Calibration (Mo 8)</li> <li>• Calibrated 3D Photogrammetric Site Meshes &amp; Point Clouds (Mo 12, paired with Harrison)</li> <li>• Automated Sensor Telemetry-to-GeoScope 3D Pipeline (Mo 16, paired with Harrison)</li> <li>• Joint Landscape Architecture &amp; Engineering 3D Modeling Workshops (Mo 18)</li> <li>• Co-Lead Author on Paper 2 (3D Digital Twins &amp; Telemetry) &amp; Co-Author on Paper 4 in 5-Paper Suite <i>Boundary: Co-leads all 3D modeling, UAV photogrammetry meshes, and telemetry engineering paired with Harrison; does not manage farmer field day recruitment (Hopper) or plant IPM SOPs (Talton).</i></li> </ul> |

| INVESTIGATOR & ROLE                                                                                                                                         | GRANT EFFORT                                                   | PRIMARY OPERATIONAL DOMAIN & ASSIGNED SCOPE                                                                                                                                                                                                                                                                                                                                                                   | INDIVIDUAL DELIVERABLES & BOUNDARY ACCOUNTABILITY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <p><b>Charles "Charlie" Hopper</b></p> <p><b>SENIOR COLLABORATOR • LIVING LAB</b></p> <p>Director, SFRIC • CAES Staff</p>                                   | <p><b>\$0 Grant Salary</b><br/>(12-Mo Administrative Host)</p> | <p><b>SFRIC Living Lab Staging &amp; Landowner Outreach:</b> Hosts and stages the Living Extension Lab across SFRIC's 100+ acre facility; directs logistics, recruitment, and community engagement for 6 regional landowner field days; leads ABCD (Asset-Based Community Development) workshops for students and farmers.</p>                                                                                | <ul style="list-style-type: none"> <li>• Facility Access Agreement &amp; Living Lab Setup (Mo 1)</li> <li>• 6 Landowner Field Demonstration Days at SFRIC (Mo 14, 18, 22, 26, 30, 34)</li> <li>• ABCD Community Leadership Training Modules (Mo 12, 24)</li> <li>• Co-Author on SEC-05 and SEC-07</li> </ul> <p><i>Boundary: Facility director &amp; community engagement; zero grant fiscal accounting or classroom grading duties.</i></p>                                                                                                                                                                                                                                                                                               |
| <p><b>Dr. Hannah Talton, Ph.D., D.P.M.</b></p> <p><b>CO-INV • EXTENSION &amp; IPM LEAD</b></p> <p>Extension Plant Pathology Specialist • Extension/CAES</p> | <p><b>Extension Effort</b><br/>+ Specialist Summer Support</p> | <p><b>Multi-Tier Scouting SOPs &amp; Extension Media:</b> Develops Standard Operating Procedures for integrated field scouting—coupling aerial drone canopy health assessments (overstory) with ground quadruped/rover inspection (understory/trunk disease); validates disease/pest diagnostic protocols; authors 4 practical, adaptable NC Cooperative Extension Technical Bulletins for small farmers.</p> | <ul style="list-style-type: none"> <li>• Integrated Drone Canopy &amp; Ground Scouting SOP Guide (Mo 15)</li> <li>• 4 NC Cooperative Extension Technical Bulletins (co-designed with growers, Mo 18, 24, 30, 36)</li> <li>• Live diagnostic field demonstrations at all 6 SFRIC field days</li> <li>• Lead Author on SEC-05 (Extension &amp; Outreach Plan)</li> </ul> <p><i>Boundary: Agronomic IPM diagnostics, plant pathology validation, and Extension grower outreach. <b>Zero programming duties:</b> all robotics coding, ROS navigation, and drone telemetry are engineered by Dr. Chakraborty and a dedicated funded CS/Engineering Student Assistant. Deliverables maintain built-in flexibility based on farmer needs.</i></p> |
| <p><b>Dr. Omoanghe S. Isikhuemhen</b></p> <p><b>SENIOR FACULTY PROJECT ADVISOR</b></p> <p>Professor of Mycology • CAES</p>                                  | <p><b>\$0 Grant Salary</b><br/>(Senior Advisory Council)</p>   | <p><b>Scientific Mentorship &amp; Research Advisory:</b> Serves as high-level scientific advisor to Dr. Felicia Anike and project leadership; advises on truffle mycorrhizae experimental design, field inoculation viability, and international agroforestry standards; conducts peer reviews of draft manuscripts.</p>                                                                                      | <ul style="list-style-type: none"> <li>• Quarterly scientific advisory reviews &amp; protocol audits (Mo 6, 12, 18, 24, 30, 36)</li> <li>• Critical pre-submission review of 2 agroforestry journal manuscripts</li> </ul> <p><i>Boundary: High-level scientific advisor; bears zero Co-PI fiscal, reporting, or day-to-day administrative burdens.</i></p>                                                                                                                                                                                                                                                                                                                                                                                |

**Peer Review Evaluation Standard (Investigator Qualifications & Management Plan):** USDA NIFA CBG peer review panels allocate **20 of 100 points** specifically to assessing whether the Project Director and co-investigators possess verified, documented qualifications directly aligned with their proposed tasks. To eliminate ambiguity, the forensic matrix below maps each investigator's terminal credentials, administrative college home, verified research/pedagogical track record, and explicit operational qualification for their assigned grant deliverable.

| INVESTIGATOR & ROLE                                                                                                                                                      | TERMINAL DEGREE & ACADEMIC HOME                                                                                                                                                          | VERIFIED DOMAIN TRACK RECORD & PRIOR HONORS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | ASSIGNED PROPOSAL SCOPE & OPERATIONAL READINESS                                                                                                                                                                                                                                                                                                                                            |
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| <b>W. Chris Harrison, MLA</b><br>Assistant Professor of Landscape Architecture<br><div>PROJECT DIRECTOR • LEAD PI • 3D MODELING CO-LEAD</div>                            | <b>Master of Landscape Architecture (MLA)</b><br>Terminal Professional Degree<br>Dept. of Natural Resources & Environmental Design (NRED)<br><b>CAES (Faculty / Program Coordinator)</b> | <ul style="list-style-type: none"><li>• Program Coordinator leading design and implementation of NC A&amp;T's new 120-credit BSLA curriculum</li><li>• Lead developer &amp; software architect of AVA (Adaptive Visualization Assistant) and GeoScope GIS spatial engines</li><li>• SITES AP accredited professional and educator; author of automated 250-point SITES v2 ecological rating algorithms</li><li>• Co-investigator on USDA NIFA Evans-Allen research projects focusing on campus digital twins and environmental data</li></ul> | <b>Project Governance, 3D Spatial Computing &amp; SITES Curriculum:</b><br>Directs overall \$750k fiscal/administrative compliance, university DOR/REEport filings; leads deployment of GeoScope/AVA spatial engines; <b>directly paired with Dr. Momtanu Chakraborty to co-lead all 3D digital twin model generation, UAV photogrammetric reconstruction, and web spatial simulation.</b> |
| <b>Dr. Momtanu Chakraborty, Ph.D.</b><br>Assistant Professor of Computational Data Science and Engineering<br><div>CO-PI • ROBOTICS, UAV &amp; 3D MODELING CO-LEAD</div> | <b>Ph.D. in Computational / Electrical &amp; Computer Engineering</b><br>Department of Computational Data Science and Engineering (McNair Hall)<br><b>College of Engineering (COE)</b>   | <ul style="list-style-type: none"><li>• Published expert in agricultural robotics, sensor telemetry, and machine vision</li><li>• Directs precision bioengineering pipelines and UAV multispectral calibration</li><li>• Published authority in machine learning, computer vision, and environmental data science</li><li>• Bridges College of Engineering (COE) data science pipelines with CAES spatial and agricultural computing</li></ul>                                                                                                | <b>3D Modeling, Sensor Telemetry &amp; Engineering Simulation: Co-leads the 3D Modeling &amp; Digital Twin Core paired directly with Project Director W. Chris Harrison;</b> engineers UAV photogrammetry-to-3D-mesh processing pipelines, point-cloud conversion, and automated telemetry ingestion into GeoScope; co-directs UAV flights with CoST.                                      |

| INVESTIGATOR & ROLE                                                                                                                                                              | TERMINAL DEGREE & ACADEMIC HOME                                                                                                                                                                                                                                         | VERIFIED DOMAIN TRACK RECORD & PRIOR HONORS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ASSIGNED PROPOSAL SCOPE & OPERATIONAL READINESS                                                                                                                                                                                                                                                                                                                                                                           |
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| <b>Dr. Jun Yang, Ph.D.</b><br>Assistant Professor of Landscape Architecture<br><div>CO-PI • AI PEDAGOGY &amp; DESIGN COGNITION LEAD</div>                                        | <b>Ph.D. in Architecture &amp; Design Research (Virginia Tech)</b><br><b>M.S. in Computer Science, AI Track (Georgia Tech)</b><br>MLA (Politecnico di Milano) • B.A. (Nankai Univ.)<br>Dept. of Natural Resources & Environmental Design (NRED) • <b>CAES (Faculty)</b> | <ul style="list-style-type: none"> <li>Nationally recognized scholar in AI design cognition and pedagogical tracking</li> <li>Multiple peer-reviewed articles in <i>Landscape Research Record</i> (CELA) and EDRA</li> <li>Expert in visual spatial synthesis, behavioral analytics, and design computing</li> <li>Lead instructor for junior design studios and master planning sequences (LDAR 345, 347, 348, 349)</li> </ul>                                                                                                                                             | <b>AI Studio Pedagogy &amp; Design Cognition Study:</b> Directs generative AI design integration in foundational studios; leads longitudinal IRB-approved student cognition study evaluating spatial synthesis velocity, mental workload, and graphic ideation with AI tools vs. traditional methods.                                                                                                                     |
| <b>Jeffrey A. Kuehner, MLA, ASLA</b><br>Full-Time Lecturer of Landscape Architecture<br><div>CO-INVESTIGATOR • SENIOR CAPSTONE, PRACTICE &amp; CONSTRUCTION DETAILING LEAD</div> | <b>Master of Landscape Architecture (MLA)</b><br>Dept. of Natural Resources & Environmental Design (NRED)<br><b>CAES (Faculty / Lecturer)</b>                                                                                                                           | <ul style="list-style-type: none"> <li>15+ years of professional environmental design, site planning, and technical studio instruction</li> <li>Co-author with W. Chris Harrison on CELA Landscape Research Record (LRR) scholarship on campus digital twins and field ground-truthing</li> <li>Lead instructor for senior capstone studios (LDAR 447/448), design proposal writing (LDAR 440), and current topics (LDAR 455)</li> <li>Specialist in low-impact development (LID), grading logic, construction documents (LDAR 249), and constructible detailing</li> </ul> | <b>Senior Capstone Practice, Professional Detailing &amp; Licensure Readiness:</b> Directs senior capstone design execution (LDAR 447/448), professional contracts/proposals (LDAR 440/442), and construction documents (LDAR 249); develops the Master Construction Detailing Exemplar Library ensuring graduating seniors master professional technical documentation, constructible detailing, and industry standards. |
| <b>Dr. Leila Hashemi Beni, Ph.D.</b><br>Associate Professor & GEMS Director<br><div>CO-PI • GEOSPATIAL LEAD</div>                                                                | <b>Ph.D. in Geomatics / Remote Sensing</b><br>Dept. of Built Environment / GEMS Institute<br><b>College of Science and Technology (CoST)</b>                                                                                                                            | <ul style="list-style-type: none"> <li>Recipient of \$5M+ in external federal research grants (NASA, NOAA, USDA, NSF)</li> <li>Director of GEMS (Geospatial and Environmental Mapping Sciences) Institute</li> <li>Internationally published authority on UAV/terrestrial LiDAR and deep learning</li> <li>Directs campus multi-sensor drone fleet and spatial high-performance computing</li> </ul>                                                                                                                                                                        | <b>LiDAR Remote Sensing &amp; Spatial Data Pipeline:</b> Directs aerial and terrestrial LiDAR data collection missions across research woodlots and SFRIC; manages point cloud classification (canopy height models, ground elevation DEMs); provides calibrated spatial data layers to the GeoScope engine.                                                                                                              |

| INVESTIGATOR & ROLE                                                                                                                      | TERMINAL DEGREE & ACADEMIC HOME                                                                                                                                                  | VERIFIED DOMAIN TRACK RECORD & PRIOR HONORS                                                                                                                                                                                                                                                                                                                                                                                                       | ASSIGNED PROPOSAL SCOPE & OPERATIONAL READINESS                                                                                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Dr. Felicia N. Anike, Ph.D.</b><br>Research Scientist / Faculty<br>• Fungal Biotechnology<br><b>CO-PI • BIOECONOMY LEAD</b>           | <b>Ph.D. in Applied Microbiology &amp; Fungal Biotechnology</b><br>Dept. of Natural Resources & Environmental Design (Carver Hall)<br><b>CAES (Faculty / Research Scientist)</b> | <ul style="list-style-type: none"> <li>• Extensive research record in fungal enzymes, mycorrhizal ecology, and agroforestry bioeconomy</li> <li>• Multi-year research collaboration with Dr. Omon Isikhuemhen on pecan truffle (<i>Tuber lyonii</i>)</li> <li>• Published in leading mycology and biotechnology journals on crop residue valorization</li> <li>• Experienced in greenhouse and field mycorrhizal inoculation protocols</li> </ul> | <b>Forest Bioeconomy &amp; Fungal Biotechnology:</b> Directs agroforestry field trial establishment, The Living Fungal Forest Laboratory, fungal inoculation protocols (pecan truffle and medicinal fungi), mycorrhizal colonization tracking, soil enzyme assays in Carver Hall labs and field plots.               |
| <b>Dr. Porché L. Spence, Ph.D.</b><br>Associate Professor of Environmental Studies<br><b>CO-PI • CONSERVATION LEAD</b>                   | <b>Ph.D. in Soil Science / Environmental Biogeochemistry</b><br>Dept. of Natural Resources & Environmental Design (Carver Hall)<br><b>CAES (Faculty)</b>                         | <ul style="list-style-type: none"> <li>• Ecological Society of America (ESA) Diversity in Ecology Awardee</li> <li>• Expert in urban watershed biogeochemistry, soil organic carbon (SOC), and nutrient cycling</li> <li>• Principal investigator on USDA-funded environmental science mentoring programs</li> <li>• Outstanding record of placing underrepresented students in federal agency internships</li> </ul>                             | <b>Conservation Credentials &amp; Soil Biogeochemistry:</b> Directs NRCS Conservation Planning Apprentice credential curriculum (USDA 9-step planning); leads lab analysis of soil health, soil organic carbon, and water quality from research woodlots; correlates ground-truth carbon with canopy remote sensing. |
| <b>Dr. Chastity Warren English, Ph.D.</b><br>Associate Professor & Program Coordinator<br><b>CO-PI • EVALUATION LEAD</b>                 | <b>Ph.D. in Agricultural and Extension Education</b><br>Dept. of Agribusiness, Applied Economics and Agriscience Education (Carver Hall 251-A)<br><b>CAES (Faculty)</b>          | <ul style="list-style-type: none"> <li>• Program Coordinator of Agricultural Education; CAES Advising Excellence Awardee</li> <li>• ACUE (Association of College and University Educators) certified master instructor</li> <li>• Proven track record serving as external evaluator on multi-million-dollar USDA grants</li> <li>• Expert in curriculum efficacy, logic models, and mixed-method evaluation frameworks</li> </ul>                 | <b>Independent Project Evaluation:</b> Directs formal, ACUE-grounded assessment plan across all teaching and extension modules; develops project Logic Model, pre/post competency rubrics, and student retention/graduation tracking; measures credential attainment across NRCS, SITES AP, and ABCD cohorts.        |
| <b>Charles "Charlie" Hopper</b><br>Director, Small Farm Research & Innovation Center<br><b>SENIOR COLLABORATOR • LIVING LAB DIRECTOR</b> | <b>B.S. / Senior Agricultural Administrator</b><br>Small Farm Research and Innovation Center (SFRIC)<br><b>CAES Staff / Center Administration (\$0 Grant Salary)</b>             | <ul style="list-style-type: none"> <li>• Directs the 100+ acre SFRIC research and demonstration facility</li> <li>• Certified practitioner of Asset-Based Community Development (ABCD) methodologies</li> <li>• Established network of 500+ small-scale, limited-resource, and minority NC farmers</li> <li>• Key leader in NC A&amp;T Cooperative Extension outreach and field day programming</li> </ul>                                        | <b>SFRIC Living Lab Staging &amp; Landowner Outreach:</b> Hosts and stages the Living Extension Lab across SFRIC's 100+ acre facility; directs logistics, recruitment, and community engagement for 6 regional landowner field days; leads ABCD community development workshops.                                     |

| INVESTIGATOR & ROLE                                                                                                                                                                | TERMINAL DEGREE & ACADEMIC HOME                                                                                                                                                                      | VERIFIED DOMAIN TRACK RECORD & PRIOR HONORS                                                                                                                                                                                                                                                                                                                                                                  | ASSIGNED PROPOSAL SCOPE & OPERATIONAL READINESS                                                                                                                                                                                                                                                                                                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Dr. Hannah Talton, Ph.D., D.P.M.</b><br/>Assistant Professor &amp; Extension Specialist (IPM / Plant Pathology)</p> <p><b>CO-INVESTIGATOR • EXTENSION &amp; IPM LEAD</b></p> | <p><b>Doctor of Plant Medicine (D.P.M.) &amp; Ph.D.</b><br/>Dept. of Agriculture &amp; Environmental Sciences / Cooperative Extension<br/><b>CAES Cooperative Extension (Faculty/Specialist)</b></p> | <ul style="list-style-type: none"> <li>• Statewide specialist in integrated pest management, disease vectors, and crop diagnostics</li> <li>• Field-scale scouting protocol design and plant disease diagnostic validation</li> <li>• Author of multiple NC Cooperative Extension grower bulletins and field guides</li> <li>• Active statewide extension workshop leader for underserved farmers</li> </ul> | <p><b>Multi-Tier Scouting SOPs &amp; Extension Media:</b> Leads agronomic disease/pest validation across aerial drone canopy scans and ground robotic understory inspection; authors 4 practical Cooperative Extension Technical Bulletins for small farmers. <i>(Supported by funded CS/Engineering Student Assistant under Dr. Chakraborty for all robot coding).</i></p> |
| <p><b>Dr. Omoanghe S. Isikhuemhen, Ph.D.</b><br/>Research Professor of Fungal Biotechnology &amp; Agroforestry</p> <p><b>SENIOR FACULTY PROJECT ADVISOR</b></p>                    | <p><b>Ph.D. in Applied Mycology &amp; Microbiology</b><br/>Dept. of Natural Resources &amp; Environmental Design (Carver Hall)<br/><b>CAES (Faculty — \$0 Grant Salary)</b></p>                      | <ul style="list-style-type: none"> <li>• Globally recognized leader in mycorrhizal truffle biotechnology and agroforestry</li> <li>• Awarded multi-million-dollar USDA, NSF, and international research grants</li> <li>• 30+ years of high-impact publications in fungal bioeconomy and forest products</li> <li>• Senior mentor to Dr. Felicia Anike and junior bioeconomy faculty</li> </ul>              | <p><b>Scientific Mentorship &amp; Research Advisory:</b> Serves as high-level scientific advisor to Dr. Felicia Anike and project leadership; advises on truffle mycorrhizae experimental design, field inoculation viability, and international agroforestry standards; conducts peer reviews of manuscripts.</p>                                                          |

06. THE PROJECT SILVA COLLABORATIVE 5-PAPER SUITE  
(CROSS-COLLEGE CO-AUTHORSHIP)

Interdisciplinary Scholarship &  
Student Mentorship

**Authentic Scholarship Strategy (Replacing Fragmented Individual Paper Promises):** Review panels view proposals promising isolated, single-investigator conference papers with deep skepticism. To model authentic interdisciplinary scholarship and maximize peer-reviewed impact, the Project SILVA team commits to **The Collaborative 5-Paper Suite**. Each paper directly addresses a core project milestone, unites co-authors from multiple departments and colleges, and pairs faculty leads with **undergraduate student co-authors**.

| PAPER #                       | STRATEGIC FOCUS & WORKING TITLE                                                                                                                                                                    | INTERDISCIPLINARY AUTHORSHIP TEAM                                                                                         | TARGET PEER-REVIEWED OUTLET                                                     | MILESTONE                                                 |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------|
| PAPER 1<br>Teaching           | AI Studio Pedagogy & Design Cognition<br>"Generative AI in Undergraduate Design Studios: Longitudinal Assessment of Spatial Cognition, Design Velocity, and Mental Workload"                       | Dr. Jun Yang (Lead)<br>W. Chris Harrison, Dr. Chastity Warren English, & Undergraduate Student Researcher                 | Landscape Research Record (CELA LRR) / EDRA Proceedings                         | Month 18<br>(Preliminary)<br>Month 32<br>(Final)          |
| PAPER 2<br>Research / Tech    | 3D Digital Twins & Multi-Sensor Telemetry<br>"Integrated 3D Photogrammetric Mesh Generation and Multi-Sensor Telemetry for Land-Grant Digital Farm Twins"                                          | W. Chris Harrison & Dr. Momtanu Chakraborty (Co-Leads)<br>Dr. Leila Hashemi Beni & Undergraduate Student Researcher       | Computers and Electronics in Agriculture (Elsevier) / Transactions of the ASABE | Month 20<br>(Architecture)<br>Month 34<br>(Validation)    |
| PAPER 3<br>Research / Bio     | Living Fungal Forest & Truffle Bioeconomy<br>"Establishing Pecan Truffle (Tuber lyonii) Agroforestry Polycultures in the Piedmont: Mycorrhizal Colonization and Soil Enzymatic Dynamics"           | Dr. Felicia N. Anike (Lead)<br>Dr. Porché L. Spence, Dr. Omoanghe S. Isikhuemhen, W. Chris Harrison, & Student Researcher | Agroforestry Systems (Springer) / Mycorrhiza                                    | Month 24<br>(Inoculation)<br>Month 36<br>(Biochemistry)   |
| PAPER 4<br>Research / Spatial | UAV LiDAR & Soil Carbon Biogeochemistry<br>"Correlating Multi-Temporal UAV LiDAR Canopy Metrics with Piedmont Soil Organic Carbon Stocks Under Agroforestry Management"                            | Dr. Leila Hashemi Beni (Lead)<br>Dr. Porché L. Spence, Dr. Momtanu Chakraborty, W. Chris Harrison, & Student Researcher   | Remote Sensing (MDPI) / Journal of Soil and Water Conservation                  | Month 22<br>(LiDAR Model)<br>Month 34<br>(Carbon Indices) |
| PAPER 5<br>Extension          | Robotics & Living Lab Demonstration<br>"Integrated Drone Canopy Mapping and Understory Robotic Scouting: An Extension Demonstration Model for Underserved Forestland and Specialty Crop Producers" | Dr. Hannah Talton (Lead)<br>Charles Hopper, W. Chris Harrison, Dr. Chastity Warren English, & Extension Agent Co-Author   | Journal of Extension (JOE) / NACTA Journal                                      | Month 26<br>(Field SOPs)<br>Month 36<br>(Impact Data)     |

 **Core Research Questions & Empirical Hypotheses Matrix (By Core Area)**

**Scientific Inquiry & Rigor Framework (Dr. Omon Research Mandate):** In accordance with Senior Faculty Advisor Dr. Omon Isikhuemhen's guidance, the \$225,000 Research Core is governed by explicit, testable Research Questions (RQs) and empirical Hypotheses (H). This formal scientific structure moves the project beyond generic descriptive activities into rigorous, reproducible inquiry across four cross-college domains.

| RESEARCH AREA & LEADERSHIP                                                                                                                                                 | CORE RESEARCH QUESTIONS (RQS)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | TESTABLE HYPOTHESES & EXPERIMENTAL METHODS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><b>1. Forest Bioeconomy &amp; Mycorrhizae</b></div> <div><b>DR. FELICIA ANIKE (LEAD)</b></div> <div>Advisory: Dr. Omon Isikhuemhen<br/>Co-PI: Dr. Porché Spence</div> | <p><b>RQ1 (Mycorrhizal Colonization &amp; Soil Health):</b> What are the survival, colonization dynamics, and soil microbiome succession rates of native pecan truffle (<i>Tuber lyonii</i>) mycelial inoculation across varied Piedmont soil associations (Cecil vs. Enon series) under managed hardwood and pecan canopy systems?</p> <p><b>RQ2 (Non-Timber Polyculture Viability):</b> How does multi-strata companion cultivation of gourmet/medicinal macrofungi (<i>Pleurotus</i> and <i>Ganoderma</i> species on forest thinnings) alter understory microclimatic moisture retention and non-timber revenue viability for small-scale woodlot owners?</p> | <p><b>Hypothesis 1 (H1):</b> Inoculated host rootzones will demonstrate a statistically significant (&gt;30%) increase in ectomycorrhizal root-tip colonization within 24 months, accompanied by measurable increases in labile soil organic carbon (SOC) fractions and glomalin-related soil proteins relative to non-inoculated controls.</p> <p><i>Methodology:</i> DNA-based fungal metabarcoding, ergosterol assays, and Carver Hall root-tip microscopy coupled with randomized field plots across SFRIC and university research woodlots.</p> <p><b>Output:</b> Paper 3 (<i>Agroforestry Systems</i>).</p> |
| <div><b>2. Spatial Computing &amp; AI Cognition</b></div> <div><b>W. CHRIS HARRISON (LEAD)</b></div> <div>Co-PI: Dr. Jun Yang<br/>Co-PI: Dr. Momtanu Chakraborty</div>     | <p><b>RQ3 (Design Cognition &amp; Spatial Fluency):</b> How does the structured integration of generative AI spatial synthesis (AVA) and real-time GIS spatial constraints (GeoScope) impact design cognition, iterative ideation velocity, and evidence-based ecological decision-making in undergraduate design studios compared to traditional CAD/manual workflows?</p> <p><b>RQ4 (Multi-Sensor 3D Mesh Reconstruction):</b> What algorithms and mesh-fusion techniques optimize the geometric fidelity of 3D digital twins when reconciling aerial UAV photogrammetry with ground-level mobile sensor captures under dense forest canopies?</p>             | <p><b>Hypothesis 2 (H2):</b> Students utilizing the integrated spatial computing suite will generate ecologically viable design alternatives 35% faster while achieving significantly higher SITES v2 sustainability baseline scores during preliminary site master planning.</p> <p><i>Methodology:</i> Longitudinal cognitive tracking, NASA-TLX workload indices, spatial protocol analysis, and point cloud registration benchmarking in the Spatial Design Center Lab.</p> <p><b>Output:</b> Papers 1 &amp; 2 (<i>CELA LRR &amp; Comp. &amp; Electronics in Ag</i>).</p>                                     |

| RESEARCH AREA & LEADERSHIP                                                                                                                                                                                            | CORE RESEARCH QUESTIONS (RQS)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | TESTABLE HYPOTHESES & EXPERIMENTAL METHODS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>3. Multi-Tier Sensing &amp; Diagnostics</b></p> <p><b>DR. HANNAH TALTON (IPM LEAD)</b></p> <p>Co-PI: Dr. Leila Hashemi Beni (UAV)<br/>Co-PI: Dr. M. Chakraborty (Robotics)<br/>Collaborator: Charles Hopper</p> | <p><b>RQ5 (Multi-Tier Canopy vs. Understory Diagnostic Complementarity):</b> What is the diagnostic complementarity between aerial drone multispectral imagery (measuring tree crown vigor and upper canopy stress) and ground quadruped sensors (inspecting trunk-level blights, root collars, and understory specialty crops) for early disease detection?</p> <p><b>RQ6 (Canopy Penetration &amp; Topographic Accuracy):</b> How accurately can multi-sensor drone LiDAR penetrate dense temperate Piedmont forest canopies to reconstruct understory topography and hydraulic flow lines for agroforestry site planning?</p> | <p><b>Hypothesis 3 (H3):</b> Coupling aerial drone canopy scouting with targeted ground quadruped understory inspection will reduce time-to-detection of foliar and systemic pathogens by at least 40% across 100-acre demonstration plots while eliminating soil compaction compared to tractor passes.</p> <p><i>Methodology:</i> Coordinated UAV flights and quadruped waypoint transects (programmed by CS student under Dr. Chakraborty) ground-truthed by Dr. Talton's pathology assays at SFRIC.</p> <p><b>Output:</b> Papers 4 &amp; 5 (<i>Remote Sensing &amp; Journal of Extension</i>).</p>                 |
| <p><b>4. Workforce Credentialing &amp; Retention</b></p> <p><b>DR. CHASTITY WARREN ENGLISH (LEAD)</b></p> <p>Co-PI: Dr. Porché Spence<br/>Lead PI: W. Chris Harrison</p>                                              | <p><b>RQ7 (Longitudinal Credentialing &amp; Professional Self-Efficacy):</b> How does the formal embedding of federal and industry micro-credentials (USDA NRCS Apprentice Conservation Planner, SITES AP, ABCD Leadership) into required studio curricula affect academic self-efficacy, retention, and career trajectory among historically underrepresented students at an 1890 Land-Grant institution?</p>                                                                                                                                                                                                                   | <p><b>Hypothesis 4 (H4):</b> Cohorts earning verified federal certifications during their undergraduate degree program will demonstrate significantly higher post-graduation placement rates in USDA agencies (NRCS, USFS, FSA) and professional planning firms within 6 months of graduation relative to historical departmental baselines.</p> <p><i>Methodology:</i> ACUE-aligned pre/post rubric assessments, career placement tracking, and longitudinal self-efficacy survey instruments administered across cohorts.</p> <p><b>Output:</b> External Project Evaluation Reports &amp; Paper 1 co-authorship.</p> |

## 07. PROPOSAL DELIVERABLES & WRITING SECTION ASSIGNMENTS

USDA NIFA Required  
Sections

| SECTION CODE | NIFA PROPOSAL COMPONENT                                                                                                | TARGET PAGES  | LEAD AUTHOR                                                                                                                | CONTRIBUTING AUTHORS                                                                                     |
|--------------|------------------------------------------------------------------------------------------------------------------------|---------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| SEC-01       | <b>Project Summary &amp; Abstract</b><br>Overview, objectives, expected outcomes                                       | 1 Page (250w) | <b>W. Chris Harrison (PI)</b>                                                                                              | All Co-PIs                                                                                               |
| SEC-02       | <b>Project Narrative: Background &amp; Need</b><br>1890 capacity gap, digital landscape tech & AI                      | 3.5 Pages     | <b>W. Chris Harrison (PI) &amp; Dr. Jun Yang (Co-PI)</b>                                                                   | Dr. Spence, Dr. English                                                                                  |
| SEC-03       | <b>Objective 1: Teaching Infrastructure</b><br>GeoScope, Design Center Lab, AI pedagogy, NRCS/SITES certs              | 4.5 Pages     | <b>W. Chris Harrison (PI) &amp; Dr. Jun Yang (Co-PI)</b>                                                                   | Jeffrey A. Kuehner, Dr. English, Dr. Spence                                                              |
| SEC-04       | <b>Objective 2: Research Plan</b><br>LiDAR, canopy carbon, 3D digital twin modeling, agroforestry bioeconomy           | 4.5 Pages     | <b>W. Chris Harrison &amp; Dr. Momtanu Chakraborty (3D Modeling Co-Leads) &amp; Dr. Felicia N. Anike (Bioeconomy Lead)</b> | Dr. Leila Hashemi Beni (LiDAR), Dr. Porché Spence, Dr. Omon Isikhuemhen (Advisor)                        |
| SEC-05       | <b>Objective 3: Extension &amp; Outreach Plan</b><br>SFRIC living lab, quad robot dog, 3D digital farm twin, workshops | 3.5 Pages     | <b>Dr. Hannah Talton (Extension Specialist)</b>                                                                            | Charlie Hopper (SFRIC Collaborator), W. Chris Harrison & Dr. Momtanu Chakraborty (3D Farm Twin Co-Leads) |
| SEC-06       | <b>Project Evaluation &amp; Assessment Plan</b><br>Logic model, pre/post testing, student metrics, cognition           | 2.0 Pages     | <b>Dr. Chastity Warren English</b>                                                                                         | Dr. Jun Yang, W. Chris Harrison                                                                          |
| SEC-07       | <b>Facilities, Equipment &amp; Other Resources</b><br>Design Center Lab Space, GEMS Inst, SFRIC acreage                | 2.0 Pages     | <b>Dr. Leila Hashemi Beni</b>                                                                                              | W. Chris Harrison & Dr. Momtanu Chakraborty (3D Computing & Twin Labs), Jeffrey A. Kuehner, C. Hopper    |
| SEC-08       | <b>Budget &amp; Budget Justification Narrative</b><br>\$750k breakdown: student support, equip, travel                 | 4.0 Pages     | <b>W. Chris Harrison (PI)</b>                                                                                              | All Co-PIs & DOR Specialist                                                                              |
| SEC-09       | <b>Data Management Plan (DMP)</b><br>Spatial repositories, 3D mesh archives, open sensor data                          | 2.0 Pages     | <b>Dr. Leila Hashemi Beni</b>                                                                                              | W. Chris Harrison & Dr. Momtanu Chakraborty (3D Models & Telemetry Archives)                             |

| SECTION CODE | NIFA PROPOSAL COMPONENT                                                                        | TARGET PAGES | LEAD AUTHOR              | CONTRIBUTING AUTHORS                      |
|--------------|------------------------------------------------------------------------------------------------|--------------|--------------------------|-------------------------------------------|
| SEC-10       | <b>Mentoring &amp; Diversity Plan</b><br>Undergraduate research stipends, HBCU career pathways | 1.5 Pages    | <b>Dr. Porché Spence</b> | Dr. Jun Yang, Dr. Chastity Warren English |

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CORE AREA 1: TEACHING (EDUCATION)

Objective 1: Spatial Digital Twin Studio, AI Pedagogy, Engineering  
Detailing & Credential Pathways

\$350,000 (46.7%) •  
Months 1-36

**Leadership:** W. Chris Harrison (Lead PI & 3D Modeling Co-Lead), Dr. Momtanu Chakraborty (Co-PI & 3D Systems Co-Lead), Dr. Jun Yang (Co-PI), Jeffrey A. Kuehner (Co-Investigator), Dr. Chastity Warren English (Co-PI), Dr. Porché Spence (Co-PI)

**Mission:** Modernize undergraduate design, environmental, and agricultural curricula by building high-performance spatial digital twin infrastructure, integrating AI design pedagogy with longitudinal cognitive metrics, embedding technical site engineering and construction detailing, and establishing verifiable industry credentials.

SOFTWARE, COMPUTING, AI & ENGINEERING STUDIO  
PEDAGOGY

- ✓ **AVA V.4 & GeoScope Enterprise Deployment:**  
Production rollout of the dual-engine spatial computing suite—integrating AVA's AI site intelligence and 250-point Sustainable SITES Initiative (SITES v2) rating engine with GeoScope's multi-layered GIS, 3DEP topography, and drone telemetry across 4 studio courses.
- ✓ **3D Photogrammetry & Digital Twin Studio Modules (Harrison & Dr. Chakraborty):** Curricular units teaching UAV photogrammetric scanning, point-cloud-to-3D-mesh processing, and interactive spatial simulation across design studios.
- ✓ **Spatial Design Center & Computing Lab Space:**  
Procurement and commissioning of 6 high-performance GPU rendering workstations, VR headsets, and drone data docks.
- ✓ **Junior Studio AI Pedagogy & Cognition Study (Dr. Yang):** Longitudinal IRB-approved tracking of student spatial synthesis, design velocity, and cognitive load with AI vs. traditional methods across junior studios (LDAR 345 Site Systems, LDAR 347 Landform & Stormwater, and LDAR 348 Master Planning).
- ✓ **Digital Site Engineering & Grading Modules (Kuehner):** Cut/fill calculation, digital terrain grading, and stormwater BMP detailing modules integrated into LDAR 345/347/348 using GeoScope 3DEP data.
- ✓ **Master Construction Detailing Exemplar Library (Kuehner):** Digital library of professional construction details (permeable paving, retaining walls, timber boardwalks, bioretention basins) meeting ASLA/CSI standards, directly feeding into **LDAR 181 (Site Materials)** and **LDAR 249 (Construction Documents)**.

CREDENTIALS, EVALUATION & DEGREE PATHWAYS

- ✓ **45+ Industry Credentials:** Student cohorts trained and certified in NRCS Conservation Planning (Dr. Spence), SITES AP, and ABCD Community Leadership.
- ✓ **ACUE-Grounded Project Evaluation (Dr. English):** Formal logic model, pre/post competency rubrics, and educational efficacy reporting.
- ✓ **Graduate Elevation Bridge:** Curricular and laboratory scaffolding laying the groundwork to establish the accredited Master of Landscape Architecture (MLA) and interdisciplinary M.S. degree.

## CORE AREA 2: RESEARCH

### Objective 2: Remote Sensing & Forest Bioeconomy Field Trials

\$225,000 (30.0%) • Months 6–36

**Leadership:** W. Chris Harrison (Lead PI & 3D Modeling Co-Lead), Dr. Momtanu Chakraborty (Co-PI & 3D Modeling Co-Lead), Dr. Felicia N. Anike (Co-PI & Bioeconomy Lead), Dr. Leila Hashemi Beni (Co-PI), Dr. Porché Spence (Co-PI); Dr. Omon Isikhuemhen (Senior Advisor)

**Mission:** Execute rigorous, publishable empirical field research quantifying 3D canopy structure, soil microbiome dynamics, carbon sequestration, and high-value non-timber forest crops (NTFPs).

#### REMOTE SENSING & SPATIAL DATA PIPELINE

- ✓ **UAV & Terrestrial LiDAR Data Archive:** High-density 3D point cloud baseline of NC A&T research woodlots and farm interfaces (Dr. Beni / GEMS & Dr. Chakraborty / COE).
- ✓ **Interactive 3D Digital Twin Research Testbed (Harrison & Dr. Chakraborty):** Interactive 3D site meshes and simulation environments (GeoScope/AVA) linking soil carbon, truffle plots, and canopy structure into navigable 3D models.
- ✓ **Canopy-Carbon Correlation Models:** Multi-sensor algorithms linking LiDAR tree architecture to ground-truth biomass and soil organic carbon (Dr. Spence).

#### BIOLOGICAL TRIALS & PUBLICATIONS

- ✓ **The Living Fungal Forest Laboratory (Dr. Anike & Dr. Isikhuemhen):** Permanent agroforestry biological field infrastructure establishing inoculated pecan truffle (*Tuber lyonii*) orchards and medicinal fungal polycultures at SFRIC and university woodlots with a 15–20+ year empirical research and economic demonstration horizon.
- ✓ **3 Peer-Reviewed Papers:** Interdisciplinary publications in high-impact journals (CELA *Landscape Research Record*, *Agroforestry Systems*, *Remote Sensing*).

## CORE AREA 3: EXTENSION & OUTREACH

### Objective 3: SFRIC Living Extension Lab & Multi-Tier Field Scouting

\$175,000 (23.3%) • Months 12–36

**Leadership:** Charles "Charlie" Hopper (SFRIC Director), Dr. Hannah Talton (Extension Specialist & IPM Lead), W. Chris Harrison (Lead PI & 3D Lead), Dr. Momtanu Chakraborty (Robotics & Computing Co-Lead), Dr. Leila Hashemi-Beni (Aerial Drone Lead)

**Mission:** Translate advanced university research into direct, field-scale demonstrations that empower small-scale, limited-resource, and minority forestland owners across North Carolina.

*Deliverable Adaptability:* Demonstration protocols, bulletins, and field modules maintain operational flexibility to adapt iteratively based on participating farmer feedback, seasonal field constraints, and available technical resources.

#### MULTI-TIER SCOUTING & DIGITAL FARM LIVING LAB

- ✓ **Integrated Aerial & Ground Scouting SOPs:** Protocols combining aerial drone canopy health with ground robotic understory/trunk disease detection (Dr. Talton, Dr. Hashemi-Beni, Dr. Chakraborty & CS Student).
- ✓ **SFRIC Parcel-Scale Digital Farm Twin (Harrison & Dr. Chakraborty):** Interactive 3D web model of the SFRIC facility demonstrating regenerative agroforestry, soil practices, and real-time scouting telemetry.

#### COMMUNITY ENGAGEMENT & EXTENSION MEDIA

- ✓ **6 Regional Landowner Field Days:** Hands-on demonstration workshops at SFRIC training 120+ woodland owners and agricultural extension agents (Hopper & Dr. Talton).
- ✓ **4 Extension Technical Bulletins:** Practical guides published via NC A&T Cooperative Extension on digital woodlot assessment and agroforestry enterprise.

09. COMPREHENSIVE \$750,000 BUDGET ARCHITECTURE & COST-REALISM ANALYSIS

NIFA Criterion 5 (15 Pts) • Cost-Realism & Feasibility

**Federal Budget Track Rationale (\$750,000 Integrated Maximum):** Under USDA NIFA 1890 CBG guidelines (*USDA-NIFA-CBGP-011698*), Single-Function projects (Teaching-Only or Research-Only) are capped at \$300,000–\$600,000, whereas **Integrated Projects (simultaneously executing Teaching, Research, and Extension) have a maximum allowable budget of \$750,000 (\$250,000/year across 36 months)**. For a multidisciplinary coalition spanning 11 investigators across three colleges (CAES, CoE, CoST), requesting the full \$750,000 is not merely permissible—it is the strategic standard expected by federal panels to demonstrate institutional readiness, cross-college commitment, and lasting Land-Grant capacity.

Strategic Cost-Realism Comparison: Why \$750,000 Makes Strategic Sense

| Budget Dimension                     | Full Capacity Build (\$750,000)<br>— Recommended                                                | Scaled Lean Build (\$550,000)                                                         | Minimum Viable Build (\$400,000)                                                         |
|--------------------------------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Multidisciplinary Faculty Buy-In     | Full participation across 11 investigators; 3 colleges funded; 1.0–2.0 summer months/yr.        | Reduced to 0.5 summer months; 3 Co-PIs uncompensated; weakens tri-college engagement. | Lead PI only + 2 faculty; CoE & CoST dropped; fails "Integrated" threshold.              |
| Student Stipends & Tuition           | 6 Funded Students/yr: 4 Undergrads (\$15/hr) + 2 Graduate Assistants (\$120k total).            | 3 Funded Students (2 Undergrads + 1 Grad); student research contact hours cut by 50%. | 1 Student worker only; zero graduate assistantships; undermines 4+1 MLA bridge.          |
| Spatial Lab Computing Infrastructure | 6 High-Performance GPU Workstations (RTX 4090 class) + VR Headset Suites (\$50k).               | 2 GPU Workstations; VR suites eliminated; lab bottleneck restricts studio classes.    | Software licenses only; relies on aging general student lab PCs; no VR or digital twins. |
| Field Research & Extension Assets    | Permanent Living Fungal Forest plot at SFRIC + UAV LiDAR sensors + 6 Field Days.                | Smaller test plot; borrowed drone sensors; 3 field days; reduced landowner reach.     | Desktop simulations only; zero field trials; zero extension demonstrations at SFRIC.     |
| Institutional Value & DOR Impact     | Maximizes NC A&T indirect cost recovery (~\$125k); directly justifies 5th faculty line for MLA. | Reduced institutional return; does not provide sufficient leverage for faculty line.  | Low institutional impact; perceived as small single-investigator teaching grant.         |

Three-Year Integrated Budget Allocation by Federal Object Class Category (\$750,000)

| Category & Cost Element                                                                                              | Year 1   | Year 2   | Year 3   | Total Request | % Total | Capacity Justification & Operational Focus                                                              |
|----------------------------------------------------------------------------------------------------------------------|----------|----------|----------|---------------|---------|---------------------------------------------------------------------------------------------------------|
| 1. Personnel: Faculty Summer Effort<br>Lead PI (Harrison: 2.0 mo/yr) & 7 Co-PIs across 3 Colleges (1.0 mo/yr pooled) | \$80,000 | \$80,000 | \$80,000 | \$240,000     | 32.0%   | Compensates Lead PI administration and ensures cross-college accountability across CAES, CoE, and CoST. |

| CATEGORY & COST ELEMENT                                                                                                                        | YEAR 1           | YEAR 2           | YEAR 3           | TOTAL REQUEST    | % TOTAL       | CAPACITY JUSTIFICATION & OPERATIONAL FOCUS                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------------|------------------|------------------|---------------|----------------------------------------------------------------------------------------------------------------------|
| <b>2. Student Research &amp; Assistantships</b><br>4 Undergrad Assistants (\$15/hr x 10 hr/wk) + 2 Graduate Research Assistants                | \$40,000         | \$40,000         | \$40,000         | \$120,000        | 16.0%         | Funds 18 student-semester of experiential training in digital twins, drone piloting, and biological assays.          |
| <b>3. Equipment &amp; Spatial Computing Lab</b><br>6 GPU Workstations (\$36k), VR Suites (\$14k), LiDAR Drones (\$55k), Robot Payloads (\$30k) | \$95,000         | \$25,000         | \$15,000         | \$135,000        | 18.0%         | Front-loaded Y1 capital infrastructure establishing the Spatial Design Center Lab and SFRIC robotic sensing.         |
| <b>4. Biological Materials &amp; Field Plots</b><br>Pecan truffle inoculant (\$22k), lab assays (\$18k), soil kits (\$10k), plot care (\$10k)  | \$25,000         | \$20,000         | \$15,000         | \$60,000         | 8.0%          | Installs permanent Living Fungal Forest orchards at SFRIC and supplies Carver Hall mycorrhizal laboratories.         |
| <b>5. Extension Demonstrations &amp; Travel</b><br>6 Field Days (\$18k), CELA/ASLA/ASABE travel (\$25k), 4 Extension Bulletins (\$12k)         | \$15,000         | \$20,000         | \$20,000         | \$55,000         | 7.3%          | Supports statewide landowner workshops at SFRIC and funds faculty and student peer-reviewed paper presentations.     |
| <b>6. Student Credentials &amp; Indirect Costs</b><br>45+ SITES AP & NRCS exam fees (\$15k) + University Indirect F&A (~\$125k)                | -\$              | \$65,000         | \$75,000         | \$140,000        | 18.7%         | Fully underwrites student certification costs and pays statutory indirect costs back to NC A&T Division of Research. |
| <b>TOTAL INTEGRATED ALLOCATION</b>                                                                                                             | <b>\$250,000</b> | <b>\$250,000</b> | <b>\$250,000</b> | <b>\$750,000</b> | <b>100.0%</b> | <b>Balanced Tri-Partite Land-Grant Distribution</b>                                                                  |

## The "Smart Capital" Compensation Framework: Preserving \$510,000 (68%) for Infrastructure & Students

CAPPED AT 32.0%  
FACULTY SALARY

**The 1/9th Summer Salary Trap & Budget Protection:** Under UNC System policy, 9-month faculty summer pay is calculated at 1/9th (11.11%) of base salary per month plus ~28% institutional fringe benefits. Allocating standard 1.0–1.5 summer months across an 11-person team would bloat personnel costs past \$460,000 (61% of the grant), completely gutting the budget for physical computing labs, student stipends, and field assets. To protect institutional capacity building, this proposal establishes a **Tiered Compensation Architecture** capping faculty summer effort at **\$240,000 total across 3 years (32.0%)**—preserving **\$510,000 (68.0%)** for high-performance GPU workstations, VR headsets, 6 funded student researchers (including a dedicated Computer Science & Engineering Student Assistant for robotics programming), living field plots, and landowner outreach.

### Multidisciplinary Faculty Compensation & Effort Tiering Matrix

| COMPENSATION TIER                    | INVESTIGATORS & UNITS                                                                                                                               | EFFORT / MECHANISM                                                                               | BUDGET RATIONALE & STRATEGIC FISCAL VALUE                                                                                                                                                                                   |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tier 1: Lead PI & Project Director   | <b>W. Chris Harrison</b><br>Assistant Professor (LA / CAES)                                                                                         | <b>1.5 – 2.0 Summer Months/yr</b><br>~\$22,000/yr (incl. fringe)                                 | Compensates overall \$750k fiscal/scientific administration, AVA/GeoScope software development, and annual USDA reporting.                                                                                                  |
| Tier 2: Junior Co-PI Leadership Core | <b>Dr. Jun Yang</b> (LA / CAES)<br><b>Dr. Momtanu Chakraborty</b> (COE)<br><b>Dr. Felicia Anike</b> (CAES)<br><b>Dr. Hannah Talton</b> (Extension)  | <b>0.75 – 1.0 Summer Month/yr</b><br>~\$9,500–\$12,000/yr each                                   | Leverages junior faculty base salary rates to secure high-output cross-college leadership across studios, UAV/robotics, and fungal labs without top-heavy salary drain.                                                     |
| Tier 3: Senior Faculty & Evaluation  | <b>Dr. Chastity Warren English</b><br>Associate Professor (Ag Ed / CAES)<br><b>Dr. Porché Spence</b> (CAES)<br><b>Dr. Leila Hashemi Beni</b> (CoST) | <b>Fixed Evaluator Contract (\$5,000/yr)</b><br><b>+ Graduate Research Assistantships (GRAs)</b> | <b>Saves \$42,000:</b> Budgeting Dr. English as a fixed Project Evaluator (\$5k/yr = \$15k total) avoids a 1-month Full Professor line (\$19k/yr = \$57k total). Dr. Beni and Dr. Spence receive funded GRAs in their labs. |
| Tier 4: Studio Detailing Lecturer    | <b>Jeffrey A. Kuehner</b><br>Full-Time Lecturer (LA / CAES)                                                                                         | <b>Summer Curriculum Stipend</b><br><b>\$4,500 / Summer</b>                                      | Dedicated curriculum development stipend for compiling the Master Construction Detailing Exemplar Library and site grading modules.                                                                                         |
| Tier 5: Institutional & Advisory     | <b>Dr. Omon Isikhuemhen</b> (CAES)<br><b>Charlie Hopper</b> (SFRIC)                                                                                 | <b>\$0 Direct Salary (In-Kind &amp; Research Plot Care)</b>                                      | Charlie Hopper participates as 12-month administrative director providing facility acreage. Dr. Isikhuemhen provides senior scientific advisory oversight.                                                                  |

**× DISQUALIFIED** **Unmanaged "Bloated Salary" Model (\$460K / 61%):** Over-allocates 1.0–1.5 summer months to 11 faculty; swallows \$460k in salaries + fringe; reduces

**✓ FUNDED** **Project SILVA "Smart Capital" Model (\$240K / 32%):** Caps faculty effort at \$240k; frees up \$510,000 to fund 6 student researchers, 6 GPU

student researchers to 1; cuts lab hardware to 2 desktop PCs; eliminates VR suites and field trials.

workstations, VR rigs, drone LiDAR, living fungal plots, 6 landowner field days, and university F&A.

## FEDERAL INDUSTRY CREDENTIAL ARCHITECTURE

### USDA NRCS Conservation Planner Curricula Designation: Apprentice Conservation Planner Role (Curriculum ID: NRCS-CP-Apprentice)

TITLE 180 NPPH  
PART 600  
COMPLIANT

**Federal Governing Standard & Land-Grant Precedent:** Under USDA NRCS General Manual Title 180, Part 409 (Conservation Planning Policy) and the National Planning Procedures Handbook (Title 180 NPPH, Part 600), the **Apprentice Conservation Planner** is the official prerequisite qualification for delivering conservation plans on American agricultural lands. Emulating the national land-grant model established by **Iowa State University's NRCS Curricula Designation**, North Carolina A&T will establish a formalized university articulation agreement where four accredited Landscape Architecture and Environmental Studies courses directly satisfy federal AgLearn training requirements.

#### Curricular Cross-Walk: Embedding Federal Requirements into NC A&T Coursework

| FEDERAL AGLEARN<br>MODULE / STANDARD              | CURRICULUM<br>ID     | EMBEDDED NC A&T<br>COURSE                                                   | COMPETENCY & DELIVERABLE<br>DEMONSTRATED                                                                                                        |
|---------------------------------------------------|----------------------|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Conservation Planning,<br>Part 1                  | NRCS-NEDC-<br>000019 | <b>LDAR 347/348 &amp; LDAR 440</b><br>LA Studio 3/4 & Professional Practice | Mastery of the USDA 9-Step Conservation Planning Process; completing producer objectives, resource inventories, and RMS alternatives.           |
| Intro to Field Office<br>Technical Guide (FOTG)   | NRCS-NEDC-<br>000149 | <b>LDAR 248 &amp; LDAR 345</b><br>Site Systems & Planting Design            | Navigating localized Section 1–5 FOTG technical standards, ecological site descriptions (ESD), and Piedmont vegetative practice specifications. |
| Cultural Resources<br>Training Series, Part 1     | NRCS-NEDC-<br>000141 | <b>LDAR 103 &amp; LDAR 440</b><br>Environment & Design; Prof. Practice      | National Historic Preservation Act (NHPA Section 106) compliance; cultural resource screening on working farm landscapes.                       |
| Environmental Evaluation<br>& Resource Assessment | NRCS-NEDC-<br>000216 | <b>LDAR 248 &amp; LDAR 345</b><br>Ecological Systems Studio & Site Systems  | Completing the NRCS-CPA-52 Environmental Evaluation worksheet; quantifying soil health, water quality, and Piedmont runoff mitigation.          |

**Direct Workforce Impact:** Upon completing this integrated sequence, **45+ graduating Aggie students earn the official USDA NRCS Apprentice Conservation Planner designation.** This directly pre-qualifies students for immediate federal entry at GS-7/GS-9 Soil Conservationist classifications within USDA NRCS, Farm Service Agency (FSA), U.S. Forest Service (USFS), North Carolina Soil & Water Conservation Districts (SWCD), and private ecological consulting firms—creating an institutionalized, high-wage career superhighway for historically underrepresented professionals.

**Chairperson Directive on Grant Writing Authenticity (Dr. Niroj Aryal):** Federal grant reviewers penalize proposals showing artificial intelligence "disconnects"—unnatural transitions, repetitive buzzwords, logical discrepancies across sections, or vague generalities. To ensure our proposal receives the highest review scores, the team will adhere to the following **Four-Point Human Scholarship & Review Protocol**:

1. Cross-Section Narrative & Numerical Cohesion

Ensure strict consistency across all components: student cohort sizes (Cohort 1: 20, Cohort 2: 25; Total: 45), exact budget allocations (\$350k Teaching, \$225k Research, \$175k Extension), timeline milestones, and credential numbers (NRCS, SITES AP, ABCD). Every deliverable listed in the narrative must appear in the evaluation matrix and budget justification.

2. Empirical Campus & Regional Grounding

Root every claim in physical NC A&T and Piedmont realities: reference Carver Hall research labs, the designated Spatial Design Center Lab space, GEMS Institute drone fleets, specific SFRIC acreage coordinates, actual Piedmont soil series (Cecil, Enon, Mecklenburg), and precise regional agroforestry partners. Avoid vague academic platitudes.

3. Human Faculty Section Ownership

Each investigator writes and certifies their specific domain section in their authentic scholarly voice: Dr. Yang on AI design pedagogy and studio cognition; Kuehner on technical site engineering and construction detailing; Dr. Anike on fungal biotechnology protocols; Harrison & Dr. Chakraborty jointly certifying all 3D modeling, UAV photogrammetric mesh reconstructions, and biological robotics interfaces; Dr. Beni on LiDAR calibration; Dr. Spence on soil biogeochemistry; Dr. English on ACUE-grounded assessment baselines; and Harrison on spatial computing architecture.

4. Verified Peer-Reviewed Literature Grounding

Every citation in the narrative must be verified from primary academic literature (CELA Landscape Research Record, Agroforestry Systems, Remote Sensing of Environment, USDA NIFA technical publications, Journal of Soil and Water Conservation). Zero AI-hallucinated or orphan citations are permitted in the final bibliography.

**Required Internal Competition Fields (Ready for Copy/Paste):**

- **Project Title:** Project SILVA: Spatial Intelligence for Living Vegetation & Agroforestry — An Integrated Land-Grant Platform for Spatial Digital Twins, Remote Sensing, Forest Bioeconomy, and Robotic Agricultural Extension
- **Grant Program:** USDA NIFA 1890 Institution Capacity Building Grants (CBG) Program
- **Project Type / Track:** Integrated Project Track (Teaching Lead + Research + Extension)
- **Requested Amount:** \$750,000 over 36 months (\$250,000/year)
- **Principal Investigator (Lead):** William Chris Harrison, MLA, Assistant Professor, Dept. of Natural Resources & Environmental Design (CAES) — *Project Director & Lead PI (3D Modeling & Spatial Computing Co-Lead)*
- **Co-Principal Investigators (Academic Faculty):**
  - Dr. Jun Yang, Assistant Professor of Landscape Architecture, Dept. of Natural Resources & Environmental Design (CAES) — *Co-PI (AI Pedagogy & Design Cognition Lead)*
  - Dr. Leila Hashemi Beni, Associate Professor & GEMS Director, Dept. of Built Environment (CoST) — *Co-PI (Geospatial Lead)*
  - Dr. Felicia N. Anike, Research Scientist / Faculty in Fungal Biotechnology, Dept. of Natural Resources & Environmental Design (CAES) — *Co-PI (Bioeconomy Lead)*
  - Dr. Momtanu Chakraborty, Assistant Professor of Computational Data Science and Engineering, Dept. of Computational Data Science and Engineering (COE) — *Co-PI (Robotics, UAV Systems & 3D Modeling Co-Lead paired with Lead PI)*
  - Dr. Porché L. Spence, Associate Professor of Environmental Studies, Dept. of Natural Resources & Environmental Design (CAES) — *Co-PI (Conservation & Credentials Lead)*
  - Dr. Chastity Warren English, Professor & Program Coordinator of Agricultural & Extension Education, Dept. of Agribusiness, Applied Economics & Agriscience Education (CAES) — *Co-PI (Project Evaluation Director)*
- **Co-Investigators & Key Teaching Faculty:**
  - Jeffrey A. Kuehner, MLA, ASLA, Full-Time Lecturer of Landscape Architecture, Dept. of Natural Resources & Environmental Design (CAES) — *Co-Investigator (Senior Capstone Studio, Professional Practice & Construction Detailing Lead)*
- **Senior Project Advisors & Institutional Collaborators:**
  - Dr. Omoanghe S. Isikhuemhen, Professor of Fungal Biotechnology & Agroforestry, Dept. of Natural Resources & Environmental Design (CAES) — *Senior Faculty Project Advisor*
  - Charles "Charlie" Hopper, Director, Small Farm Research and Innovation Center (SFRIC) — *Senior Institutional Collaborator & Living Lab Facility Host (Administrative/Staff line — no grant salary requested)*
  - Dr. Hannah Talton, Assistant Professor - Extension Plant Pathology Specialist, Cooperative Extension / CAES — *Extension Co-Investigator (Robotics Scouting & Agricultural Outreach Lead)*
- **Participating Colleges:** College of Agriculture and Environmental Sciences (CAES), College of Science and Technology (CoST), College of Engineering (COE), Cooperative Extension
- **Official Capacity Building & Institutional Transformation Statement (Copy/Paste for Portal Abstract):**

"The Project SILVA establishes permanent institutional capacity across North Carolina A&T State University through five enduring structural transformations: (1) **Physical & Digital Infrastructure:** Commissioning the permanent Spatial Design Center & Collaborative Computing Lab Space equipped with 6 high-performance GPU workstations, immersive VR suites, and the enterprise rollout of the **SILVA Spatial Computing Ecosystem**—advancing the **AVA (Adaptive Visualization Assistant)** AI site design and SITES v2 ecological rating engine integrated with **GeoScope** GIS and interactive 3D digital twin models co-developed across computational data science and engineering; (2) **Permanent Biological Research Infrastructure:** Establishing **The Living Fungal Forest Laboratory** across NC A&T research woodlots and the SFRIC facility, creating 15–20+ year empirical research plots for pecan truffle (*Tuber lyonii*) mycorrhizae and non-timber forest crop bioeconomies; (3) **Accredited Graduate Degree Scaffolding:** Constructing the essential technical, laboratory, and curricular foundation to establish North Carolina A&T's first professional Master of Landscape Architecture (MLA)—laying the institutional groundwork for the university to become the preeminent HBCU for licensed landscape architecture—and a Multidisciplinary M.S. in Applied Environmental Planning & Geospatial Science jointly administered across CAES, CoST, and COE; (4) **Human Capital & Credentialing:** Graduating 45+ historically underrepresented students with verified national credentials (USDA NRCS Apprentice Conservation Planner (NRCS-CP-Apprentice), SITES AP, ABCD Leadership); and (5) **Living Extension Lab:** Permanently upgrading the 100+

acre SFRIC facility into a smart-sensing living laboratory featuring autonomous quadruped robotics and 3D digital farm twins for underserved North Carolina landowners."